

Deep learning linking mechanistic models to single-cell transcriptomics data reveals transcriptional bursting in response to DNA damage

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This study presents DeepTX, a **valuable** methodological tool that integrates mechanistic stochastic models with single-cell RNA sequencing data to infer transcriptional burst kinetics at genome scale. The approach is broadly applicable and of interest to subfields such as systems biology, bioinformatics, and gene regulation. The evidence supporting the findings is **solid**, with appropriate validation on synthetic data and thoughtful discussion of limitations related to identifiability and model assumptions.

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Abstract

Cells must adopt flexible regulatory strategies to make decisions regarding their fate, including differentiation, apoptosis, or survival in the face of various external stimuli. One key cellular strategy that enables these functions is stochastic gene expression programs. However, understanding how transcriptional bursting, and consequently, cell fate, responds to DNA damage on a genome-wide scale poses a challenge. In this study, we propose an interpretable and scalable inference framework, DeepTX, that leverages deep learning methods to connect mechanistic models and scRNA-seq data, thereby revealing genome-wide transcriptional burst kinetics. This framework enables rapid and accurate solutions to transcription models and the inference of transcriptional burst kinetics from scRNA-seq data. Applying this framework to several scRNA-seq datasets of DNA-damaging drug treatments, we observed that fluctuations in transcriptional bursting induced by different drugs could lead to distinct fate decisions: IdU treatment induces differentiation in mouse embryonic stem cells by increasing the burst size of gene expression, while 5FU treatment with low and high dose increases the burst frequency of gene expression to induce cell apoptosis and survival in human colon cancer cells. Together, these results show that DeepTX can be used to

analyze single-cell transcriptomics data and can provide mechanistic insights into cell fate decisions.

Introduction

Cells must employ flexible regulatory strategies to determine their fate, including cell differentiation, survival, or apoptosis, in response to diverse external stimuli (López-Maury L et al., 2008 [↗](#); Stadhouder R et al., 2019 [↗](#)). One key cellular strategy that enables both beneficial and detrimental functions is the stochastic nature of gene expression programs, arising from intricate biochemical processes and termed “noise” (Raj A & van Oudenaarden A, 2008 [↗](#); Balázs G et al., 2011 [↗](#); Losick R & Desplan C, 2008 [↗](#)). Noise can arise at many steps—from stochastic transcription factor binding and chromatin remodeling, to variability in RNA transcription and protein translation. Among these diverse sources, transcriptional noise sits at the very start of the central dogma, converting upstream molecular fluctuations into the mRNA outputs that drive all downstream regulatory and functional processes. Transcriptional noise originates from a discontinuous pattern, wherein mRNA production occurs through alternating active and inactive gene states (Eling N et al., 2019 [↗](#); Tunnaclyffe E & Chubb JR, 2020 [↗](#)). This phenomenon, known as “transcriptional bursting”, has been observed in numerous experiments across both prokaryotic and eukaryotic cells (Golding I et al., 2005 [↗](#); Chubb JR et al., 2006 [↗](#); Larson DR et al., 2011 [↗](#)). The burst kinetics, characterized by burst frequency and burst size, describe the stochastic behavior of molecular interactions at the gene expression process, offering valuable insights into how cells encode and leverage variability. However, it remains an unresolved question how cells make fate decisions in response to external signals through genome-wide regulation of transcriptional burst mechanisms (Lammers NC et al., 2020 [↗](#); Rodríguez J & Larson DR, 2020 [↗](#)).

DNA damage is a crucial and prevalent factor contributing to cellular responses, inducing varying degrees of cytotoxicity and thereby influencing diverse cell fate decisions (Krenning L et al., 2019 [↗](#); Su TT, 2006 [↗](#); Hafner A et al., 2019 [↗](#)). For instance, low cytotoxicity limits cell proliferation (van den Berg J et al., 2019 [↗](#); Arora M et al., 2017 [↗](#); Barr AR et al., 2017 [↗](#); Deng Z et al., 2023 [↗](#)), whereas heightened cytotoxicity prompts cell differentiation and senescence (Müllers E et al., 2014 [↗](#); Feringa FM et al., 2018 [↗](#); Toledo LI et al., 2008 [↗](#); Zhao Y et al., 2023 [↗](#)). Furthermore, extreme scenarios of high cytotoxicity can cause cells to engage in the apoptotic process (Yousefzadeh M et al., 2021 [↗](#); Carneiro BA & El-Deiry WS, 2020 [↗](#); Roos WP & Kaina B, 2013 [↗](#); Zheng P et al., 2018 [↗](#)). Experiments have shown that DNA damage can disrupt the gene transcription process (Lans H et al., 2019 [↗](#)). Specifically, DNA damage slows down the movement of RNA Pol II along the DNA strand (Muñoz MJ et al., 2009 [↗](#)). When more severe DNA damage is encountered, RNA Pol II exhibits a sliding pause behavior for DNA repair before resuming transcription (**Figure 1A** [↗](#)) (Gregersen LH & Svejstrup JQ, 2018 [↗](#); Geijer ME & Marteijn JA, 2018 [↗](#); Giono LE et al., 2016 [↗](#)). These changes in RNA Pol II movement behavior would further reflect the cellular regulation of gene expression, particularly burst kinetics (Friedrich D et al., 2019 [↗](#); Calia GP et al., 2023 [↗](#)). A recent study has shown that cellular responses to DNA damage involve the regulation of burst frequency (BF) on specific genes, rather than a genome-wide conclusion (Friedrich D et al., 2019 [↗](#)). They have demonstrated that cells respond to DNA damage by increasing gene expression noise, but they do not delve into the topic of transcriptional bursts (Calia GP et al., 2023 [↗](#); Desai RV et al., 2021 [↗](#)). Therefore, the question of how DNA damage causes cells to regulate transcriptional burst kinetics on a genome-wide scale remains unresolved.

To understand the dynamics of transcriptional bursts, ideally, one needs to observe the fluctuations of gene expression over a continuous time interval, such as smFISH, scNT-seq, and HT-smFISH (Femino AM et al., 1998 [↗](#); Qiu Q et al., 2020 [↗](#); Safieddine A et al., 2023 [↗](#)). However, most existing techniques can only observe a limited number of genes, making their application to genome-wide studies challenging. Fortunately, single-cell RNA sequencing (scRNA-seq) technology provides excellent opportunities to explore this question, emerging as the leading method for

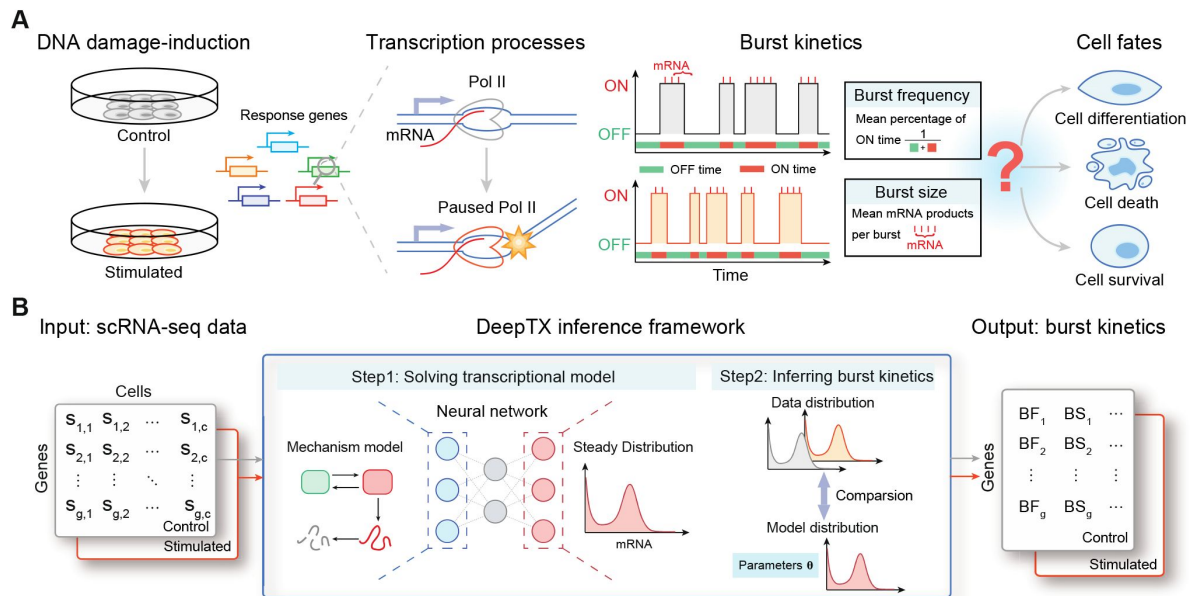


Figure 1.

Overview of DeepTX framework.

(A) Cells stimulated by DNA damaging drugs cause damage to the DNA double strands, which will slow down or even stop the movement of RNA Pol II on the DNA double strands. Changes in the state of RNA Pol II movement will lead to changes in the kinetics of gene expression bursts (including burst frequency and burst size), which will affect cell fate decisions such as apoptosis, differentiation, and survival. **(B)** The input to the DeepTX framework is scRNA-seq data, and the output is the burst kinetics corresponding to each gene. The core of the DeepTX inference framework is a hierarchical model, which is a mixture of the stationary distribution of the mechanism model solved by the neural network (Step 1) and the binomial distribution followed by the sequencing process. This hierarchical model is used to infer the out dynamics parameters corresponding to the data distribution (Step 2).

genome-wide mRNA measurements and revealing gene expression noise within individual cells (Tanay A & Regev A, 2017 [DOI](#); Picelli S et al., 2013 [DOI](#); Zheng GX et al., 2017 [DOI](#)). Many studies have employed scRNA-seq to elucidate the diverse sources of gene expression noise and the fundamental principles of transcriptional dynamics on a genome-wide scale (Eling N et al., 2019 [DOI](#); Faure AJ et al., 2017 [DOI](#); Morgan MD & Marioni JC, 2018 [DOI](#); Ochiai H et al., 2020 [DOI](#)). However, scRNA-seq data are static snapshot measurements. To recover the underlying dynamic transcriptional burst kinetics, a mathematical approach is needed to model the dynamic gene expression stochastic processes by regarding the scRNA-seq data as a steady-state result of the dynamic model (Larsson AJM et al., 2019 [DOI](#); Luo S et al., 2023 [DOI](#); Luo S et al., 2023 [DOI](#)).

For mathematical modeling, the gene expression model for DNA damage ought to encompass more sophisticated generalized properties compared to the traditional telegraph model, which is a common model describing gene expression burst kinetics by genes switching stochastically between active (ON) and inactive (OFF) states with single-step process (exponentially distributed waiting times) (Stumpf PS et al., 2017 [DOI](#); Voss TC & Hager GL, 2014 [DOI](#)). However, the presence of DNA damage necessitates modeling the transcriptional process as a multi-step process, rather than a single-step process, to capture the additional complexity introduced by the damage (Singh A et al., 2013 [DOI](#); Cavallaro M et al., 2021 [DOI](#)). Many efforts in this field are addressing the challenges of describing this multi-step process of gene expression in an interpretable and tractable model. One common modeling approach is to introduce multiple intermediate states between inactivate and activate states (Zhang J et al., 2012 [DOI](#); Zhang J & Zhou T, 2014 [DOI](#); Zhou T & Zhang J, 2012 [DOI](#)). Although this multi-state model can fit the experimental data better, it is hard to rationalize the state numbers and parameters (Desai RV et al., 2021 [DOI](#); Rodriguez J et al., 2019 [DOI](#); Zoller B et al., 2015 [DOI](#)). Another alternative modeling approach is direct non-Markovian modeling for non-exponential waiting times in gene state, which maps the multiple parameters from multi-state models to a small number of interpretable parameters that are easily observed experimentally (Daigle BJ, Jr. et al., 2015 [DOI](#); Kumar N et al., 2015 [DOI](#); Schwabe A et al., 2012 [DOI](#); Stinchcombe AR et al., 2012 [DOI](#); Zhang J & Zhou T, 2019 [DOI](#); Zhang JJ & Zhou TS, 2019 [DOI](#)). However, the ensuing difficulty is solving the analytic solution of the steady distribution from a non-Markovian model, which may require some numerical simulation methods that are computationally resource-intensive and time-consuming, such as Monte Carlo methods (Sisson SA et al., 2007 [DOI](#)).

Statistical inference, particularly in recovering burst kinetic parameters from genome-wide scRNA-seq data, necessitates efficient and scalable inference algorithms (Gómez-Schiavon M et al., 2017 [DOI](#)). Therefore, the stochastic dynamical system must be quickly solved as parameters are continuously updated throughout the inference process. Deep learning exhibits a broad array of applications as a contemporary method for addressing complex systems (Jiang Q et al., 2021 [DOI](#); Wang S et al., 2019 [DOI](#); Michoski C et al., 2020 [DOI](#)). For example, some studies employed neural networks to establish the mapping relationship between a wide range of parameters in the gene expression model and the corresponding pre-simulated steady distribution (**Figure 2B**) (Jiang Q et al., 2021 [DOI](#); Wang S et al., 2019 [DOI](#); Davis CN et al., 2020 [DOI](#); Tang Y et al., 2023 [DOI](#)). This approach, based on neural networks, has been utilized for parameter inference in both deterministic and stochastic models (Jiang Q et al., 2021 [DOI](#); Gaskin T et al., 2023 [DOI](#); Sukys A et al., 2022 [DOI](#); Tang W et al., 2023 [DOI](#)).

In this study, we introduce DeepTX, a deep learning inference framework that integrates mechanistic models and deep learning methods to elucidate the genome-wide regulation of transcriptional bursts in DNA damage response using scRNA-seq data. The DeepTX framework comprises two modules: DeepTXsolver and DeepTXinferer. Specifically, DeepTX initially employs the DeepTXsolver module to efficiently solve complex gene expression dynamics models using neural network architectures. It subsequently utilizes the DeepTXinferer module to accurately infer potential transcriptional burst kinetic parameters using Bayesian methods. DeepTX demonstrates good performance on synthetic datasets. Furthermore, to investigate genome-wide regulation of transcriptional bursts in response to DNA damage, we applied the DeepTX

A Step1: Solving TX model by deep learning

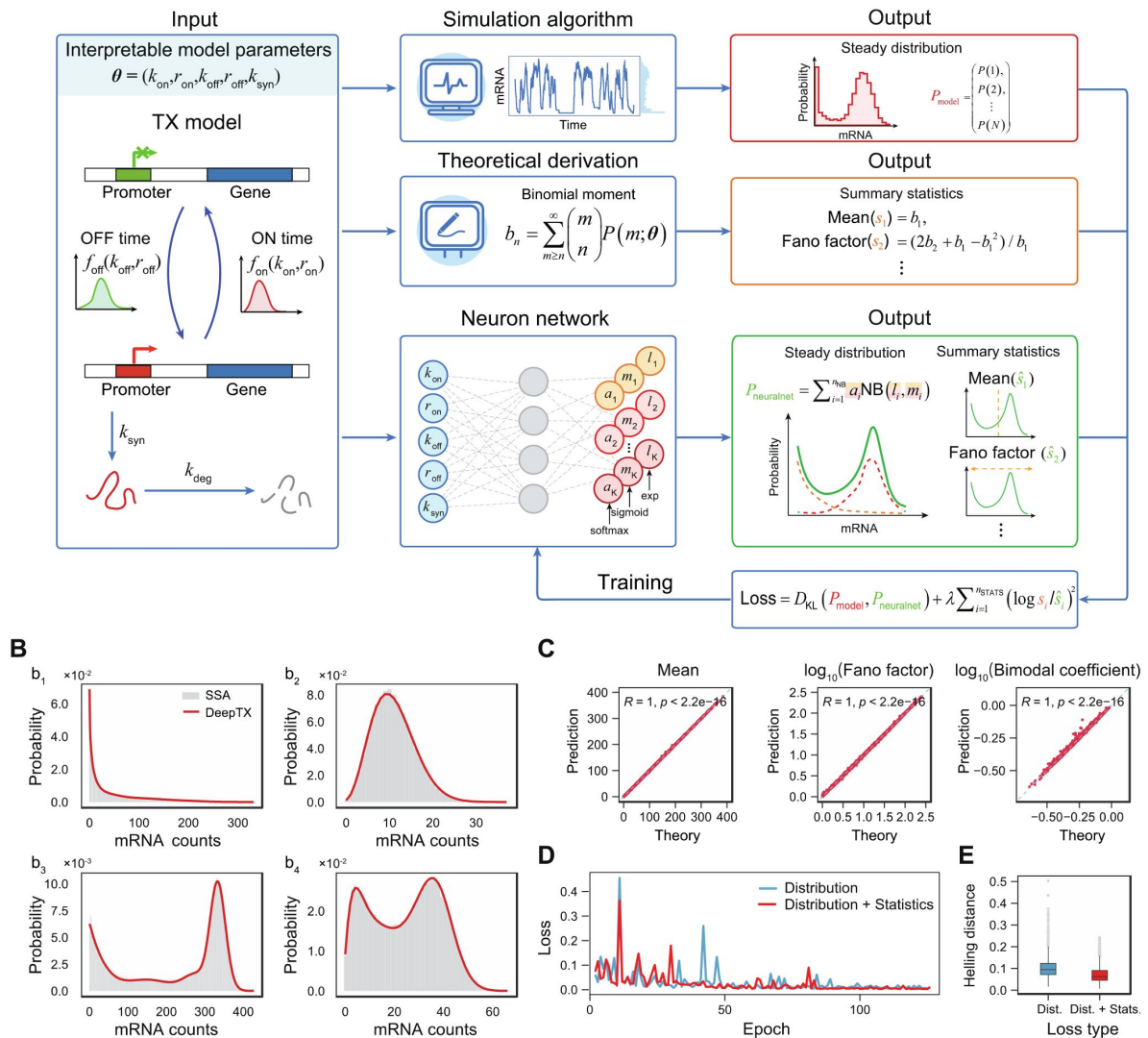


Figure 2.

Solving TXmodel using deep learning.

(A) With the TXmodel and corresponding model parameters θ , we obtain a steady distribution using the SSA simulation and the moment statistics employing binomial moment theory. These results are compared with the distribution and moment statistics of the neural network's output to calculate the neural network's loss value. The parameters of the neural network are optimized using gradient descent until convergence is achieved, indicating attainment of optimal parameters. (B) Comparison between SSA results and DeepTX prediction results for four sets of test parameters. The gray histograms show the discrete mRNA distributions simulated by SSA. The red line connects the predicted probabilities at each mRNA count from DeepTX, providing a visual representation of the discrete predicted distribution. (C) Verification results of the moment statistics predicted by DeepTX and the true moments on a test set containing 1000 elements. (D) The loss curve during DeepTX training. The blue curve represents that the loss function is composed of KL divergence of distributions, and the red curve represents that the loss function is composed of KL divergence of distributions and statistics. (E) Box plots of the Hellinger distance between the true distribution and the predicted distribution by DeepTX of different loss types on the test set.

framework to three DNA damage-related scRNA-seq datasets representing cell differentiation, apoptosis, and cell survival. As a result, firstly, we found that the fate decision of mouse embryonic stem cells (mESCs) to undergo cell differentiation in response to DNA damage caused by 5'-iodo-2'-deoxyuridine (IdU) compounds may be driven by enhancing burst size (BS enhancer) of genes that delay the cell mitosis phase transition cycle, thereby influencing cell reprogramming and differentiation. Secondly, the fate decision of colorectal cancer cells to cell apoptosis in response to DNA damage induced by low dose 5-fluorouracil (5FU) drug may be driven by enhancing burst frequency (BF enhancer) of genes that increase intracellular oxidative stress. Thirdly, the fate decision of colorectal cancer cells to cell survival in response to DNA damage induced by high dose 5FU drug may be driven by BF enhancer genes that induce telomerase extension to slow down oxidative stress and help cells to develop drug resistance. In conclusion, DeepTX is a computational framework with utility and extensibility to infer transcriptional dynamics from scRNA-seq data.

Results

Overview of DeepTX framework

To understand how cell influences underlying transcriptional mechanisms in response to DNA damage, as described in the introduction section (**Figure 1A** [↗](#)), we propose an effective computational inference framework, referred to as DeepTX (**Figure 1B** [↗](#)). DeepTX takes as input a set of scRNA-seq datasets and then outputs genome-wide burst kinetics parameters with the benefit from deep learning methods (**Figure 1B** [↗](#)). The inference process of the DeepTX framework is composed of two crucial modules, DeepTXsolver and DeepTXinferer. The first module, DeepTXsolver, is to solve for the stationary distribution of a gene expression dynamic model (**Figure 1B middle** [↗](#)). A neural network is constructed in DeepTXsolver to approximate the mapping from model parameters to the corresponding stationary distribution. As in many biological contexts, there is a need to model dynamics that describe more realistic gene expression processes that are difficult to solve. For instance, gene expression under DNA damage is better represented by multi-step transcriptional models, which more faithfully reflect the underlying regulatory complexity. Moreover, sequencing noise should be integrated into the modeling framework (see Methods). The second module, DeepTXinferer, infers the burst kinetics parameter (**Figure 1B middle** [↗](#)). Using a trained neural network from DeepTXsolver, we can easily and quickly obtain the stationary distribution (called model distribution) of any parameter within a reasonable parameter space. DeepTXinferer compares the model distribution to the stationary distribution of each gene in the scRNA-seq data (called data distribution) as the parameters are iterated until convergence and utilize the Bayesian inference to output the posterior distribution of burst kinetic parameter (**Figure 1B right** [↗](#)). We will describe these two modules in detail in the following two sections.

DeepTX solves the transcriptional model using deep learning

To enable the recovery of kinetic information from static scRNA-seq data, we first performed rational mechanistic modeling of gene expression processes with DNA damage. The gene transcription burst process is often characterized using the traditional two-state model ([Singh A et al., 2013](#) [↗](#); [Cavallaro M et al., 2021](#) [↗](#)), which assumes that genes switch randomly between OFF and ON states with exponentially distributed waiting times for each state. However, the gene expression process under conditions of DNA damage can result in slowing or even stopping the RNA pol II movement ([Lans H et al., 2019](#) [↗](#)). Further, DNA damage will cause many macromolecules to be recruited for damage repair, and this process will affect the spatially localized behavior of the promoter ([Lans H et al., 2019](#) [↗](#)), causing promoter inactivation and activation to become a complex multi-step process. We therefore used a generalized model we developed previously ([Luo S et al., 2023](#) [↗](#)), called TXmodel here, which extended the state waiting time to an arbitrary distribution, i.e., the gene expression system is a non-Markovian system

(Figure 2A and see Methods 1.2). More specifically, the waiting times for the transitions between the OFF and ON states are represented by two random variables, assumed to follow the arbitrary distributions $f_{\text{off}}(k_{\text{off}}, r_{\text{off}})$ and $f_{\text{on}}(k_{\text{on}}, r_{\text{on}})$, respectively. Additionally, we assumed that the rates of mRNA synthesis and degradation are constants, i.e., the waiting times for mRNA transcription and degradation follow exponential distributions with rate parameters k_{syn} and k_{deg} , respectively (Figure 2A left). Nevertheless, this non-Markovian gene expression system is difficult to solve stationary distribution analytically, while numerical solution methods are time-consuming.

For that reason, we utilized a deep learning approach, which has been demonstrated to be effective in solving stochastic systems (Gupta A et al., 2021), to construct a mapping from the parameter of the mechanistic TXmodel to its corresponding stationary distribution. More specifically, we aimed to train a fully connected neural network referred as DeepTXsolver, whose inputs are the parameters of the TXmodel θ and whose output is a stationary distribution parameterized by a mixed negative binomial distribution $P_{\text{neuralnet}}$, which has good performance in approximating the solution of the chemical master equation (Perez-Carrasco R et al., 2020; Öcal K et al., 2022) (Figure 2A, Supplementary Figure S1 and see Methods 1.3). To generate effective training sets for DeepTXsolver, we first generated a large number of parameter sets within a reasonable range of parameter space by using Sobol sampling (see Methods 1.3) (Sobol IM, 1967). Subsequently, for each parameter set, we employed a modified stochastic simulation algorithm (SSA) for non-Markovian TXmodels to generate numerous samples (Figure 2A and see Methods 1.3). The resulting stationary distributions $P_{\text{simulation}}$ served as training labels. Additionally, we demonstrated that the arbitrary order binomial moments of the TXmodel can be analytically solved iteratively within the context of queueing theory (Zhang Z et al., 2021; Zhang J et al., 2024). This enables the straightforward computation of significant summary statistics, including mean and Fano factor (Figure 2A, see Methods 1.2, and Supplementary Text 1.2). Thus, our loss function comprises two primary components: (i) the Kullback-Leibler (KL) divergence between the predicted distribution generated by the neural network $P_{\text{neuralnet}}$ and the labeled distribution generated by the SSA $P_{\text{simulation}}$, which is a widely used metric for quantifying the difference between two probability distributions; (ii) the logarithmic error of the predicted summary statistics computed by the neural network $\hat{s}$ and the labeled summary statistics s computed by theoretical derivation (Figure 2A and see Methods 1.3). Furthermore, we compared various optimizers and hyperparameters of model architectures to determine the best predicting performance (see Methods 1.3 and Supplementary Figure S2).

Consequently, the loss function of DeepTXsolver converges effectively during the training process. We utilized DeepTXsolver to predict the stationary distribution using parameters from the test set. We observed that it effectively fits all four representative distributions from the TXmodel, encompassing unimodal distribution at zero point, unimodal distribution at non-zero point, bimodal distribution with one peak at zero point, and bimodal distribution with both peaks at non-zero point (Figure 2B). These distributions have been extensively linked to cell fate decisions in numerous experiments (Gupta PB et al., 2011; Cohen AA et al., 2008; Bessarabova M et al., 2010). Moreover, we demonstrated that DeepTXsolver accurately predicted crucial distribution properties, including mean, Fano factor, and bimodal coefficient, with a high correlation between theoretical and predicted summary statistics ($r > 0.99$, $p\text{-value} < 2.2 \times 10^{-16}$ of $t\text{-test}$ Figure 2C). It is noteworthy that, to assess the impact of the presence or absence of summary statistics on DeepTXsolver, we performed separate experiments using different loss functions. We demonstrated that incorporating summary statistics into the model led to quicker convergence of DeepTXsolver on the training set (Figure 2D) and yielded more robust and accurate predictions on the test set (Figure 2E).

Overall, DeepTXsolver effectively establishes mappings from parameters to stationary distributions. This approach circumvents the high computational resource requirements of classical simulation algorithms and ensures the algorithm's efficiency and scalability in subsequent statistical inference, particularly in genome-wide inference.

DeepTX infers genome-wide transcriptional kinetics from scRNA-seq data

Having obtained the trained neural network model, we can rapidly compute the stationary distributions of any TXmodel parameters. These distributions represent the outcomes of the underlying true gene expression process, referred to as the “true distribution” P_{model} (Figure 3A). However, during the inference process, the gene expression distributions measured by the scRNA-seq data we utilized, termed the “observed distribution of the data” P_{data} , were subject to noise, including errors stemming from the measurement technique (Sarkar A & Stephens M, 2021) (Figure 3A). Thus, we introduced a mechanistic hierarchical model to bridge the gap between the true and observed distributions. This hierarchical model considers that the observed data results from the interplay between the gene expression process and the measurement process. It combines the two distributions using a convolutional form

$P_{\text{data}}(Y = y) = \int_0^{\infty} P_{\text{measure}}(y|x) P_{\text{model}}(x) dx$ (see Methods 1.4). $P_{\text{model}}(x)$ represents the mixed negative binomial distribution approximated, whereas for $P_{\text{measure}}(y|x)$, we opted for the binomial distribution. This choice is informed by the fact that it can approximate hypergeometric distributions representing sequencing sampling without replacement when the sample size is sufficiently large. Consequently, the final distribution can maintain the form of a mixed negative binomial distribution (Supplementary Note 1.4), denoted as the observed distribution of the model P_{data} (Figure 3A).

Leveraging the hierarchical model, we can construct a scalable and interpretable module called DeepTXinferer, facilitating genome-wide inference of transcriptional burst kinetics from scRNA-seq data. First, we input the prior distribution of mechanistic model parameters and a set of scRNA-seq data, and we can get the observed distribution of the model. P_{obsmodel} (derived successively from the trained neural network and the hierarchical model) and the observed distribution of the data P_{obsdata} , respectively (Figure 3A). Second, we used the Hellinger distance $L(\theta) = D(P_{\text{obsmodel}}, P_{\text{obsdata}})$ to measure the difference between P_{obsmodel} and P_{obsdata} (see Methods 1.4). It’s worth noting that the loss function remains solvable via gradients, despite the inclusion of complex computations, such as neural networks and hierarchical models. Specifically, to avoid the gradient-based optimization process from getting trapped in local optima, we employ a black-box optimization approach to select the initial point for parameter inference (Das S & Suganthan PN, 2010). Consequently, we utilize a gradient-based optimizer to update the iterative parameters until convergence. Lastly, we employ a method based on loss potentials, which computes the posterior distribution of the parameters during the optimization of the loss function (Gaskin T et al., 2023) (Figure 3A, see Methods 1.4). Subsequently, this process yields results of the transcriptional burst kinetics derived from the estimated parameters of the mechanistic model.

We assessed the effectiveness of DeepTXinferer by conducting validation on synthetic data (see Methods 1.4). Synthetic samples were generated for each parameter set, and our inference algorithm was then applied to derive the output transcriptional burst kinetics (see Methods 1.4). As a result, the estimation of transcriptional burst kinetics demonstrated high accuracy and strong correlation across most parameter regions ($r > 0.99$, $p\text{-value} < 2.2 \times 10^{-16}$ of t-test Supplementary Figure S3b, c), particularly within the region of fully expressed mRNA abundance (Figure 3B). Moreover, the estimated posterior distribution of each parameter suggests that the peak value closely approximates the true value (Figure 3C-E). Additionally, we verified the robustness of our algorithm concerning the number of samples, observing that higher sample numbers corresponded to increased accuracy in estimation (Supplementary Figure 3d, e). Compared to previous work (Luo S et al., 2023), DeepTX has achieved improvements in inference accuracy and computational efficiency (Supplementary Figure S4). In summary, our inference algorithm exhibited strong performance on synthetic data, affirming its reliability for application to real-world data in inferring transcriptional bursts.

A Step2: Inferring burst kinetics from scRNA-seq data

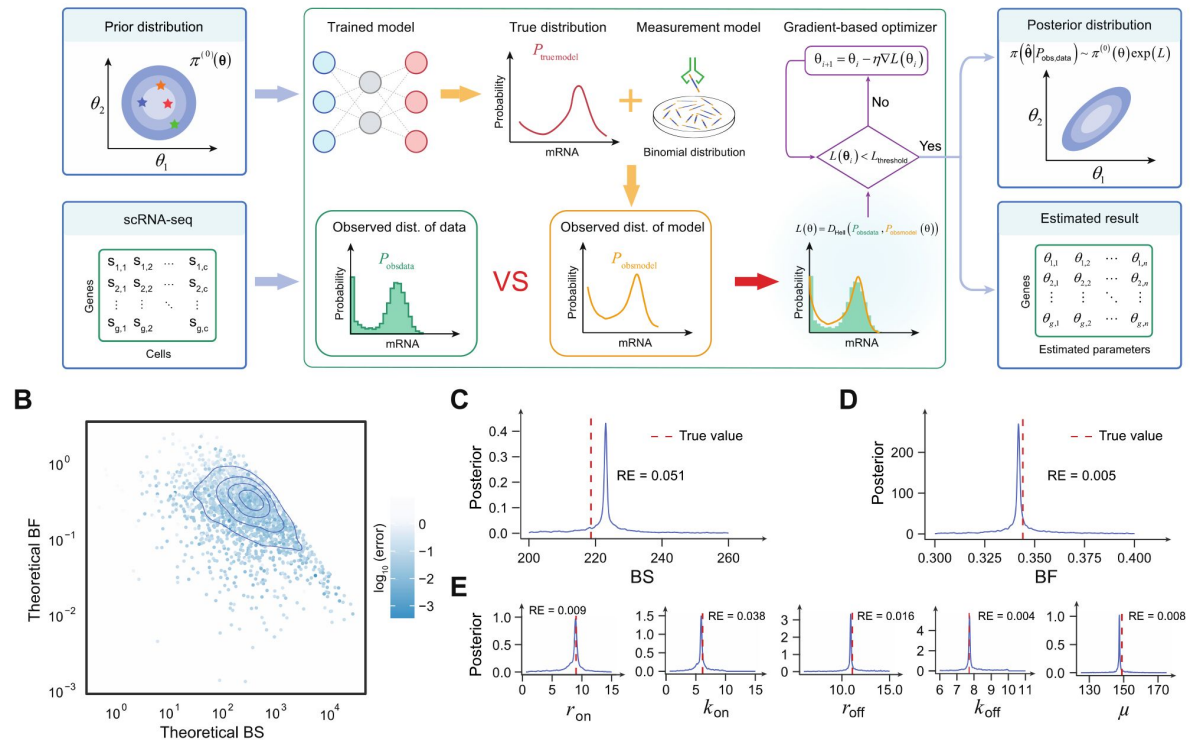


Figure 3.

Inferring TXmodel from scRNA-seq data.

(A) The dynamic parameters are sampled from a given prior distribution, and as input to the neural network, the solution of the corresponding dynamic model can be obtained by using this parameter. The solution of the dynamic model is mixed with the binomial distribution to obtain the model observed distribution. Loss values were obtained using the observed distribution of the data obtained from the scRNA-seq data compared to the observed distribution of the model. The loss values are optimized until convergence using gradient descent to obtain the parameters of the mechanism model and the posterior distribution of its parameters. **(B)** The scatters represent the real BS and BF values of the SSA synthetic set parameters, and the depth of the color represents the error between the inferred BS (BF) and the true BS (BF). **(C, D)** The blue solid line represents the edge density of the burst kinetics of the model, and the red dashed line represents the true value. **(E)** The marginal density for the five parameters of the model, where the red dashed line represents the true value. The relative error (RE) quantifies the discrepancy between the true parameter and the peak of the inferred posterior distribution.

In the subsequent sections, we will employ the DeepTXinferer to explore how cells determine diverse fates in response to DNA damage stimuli through genome-wide regulation of transcriptional bursting. We analyzed three sets of scRNA-seq data representing distinct cell fates: cell differentiation, apoptosis, and survival. The first set of data is a mutually controlled scRNA-seq dataset of mESC cell differentiation induced by IdU drug treatment. This data set contains 12481 genes of transcriptomes from 812 cells and 13780 genes of transcriptomes from 744 cells, respectively. The second set of data is a mutually controlled scRNA-seq dataset of human colon cancer cells treated with low-dose 5FU drug that causes apoptosis. This data set contains 8534 genes of transcriptomes from 1673 cells and 7077 genes of transcriptomes from 632 cells, respectively. The third set of data is a mutually controlled scRNA-seq dataset of human colon cancer cells in which high-dose 5FU drug treatment resulted in cell-resistant survival. This data set contains 8534 genes of transcriptomes from 1673 cells and 6661 genes of transcriptomes from 619 cells, respectively.

DeepTX reveals that the BS enhancer induced by IdU treatment promotes cell differentiation

The administration of the IdU drug, a thymine analogue, influences the gene expression process within cells, subsequently impacting cell differentiation (Desai RV et al., 2021 [DOI](#); Li C et al., 2024 [DOI](#)). The perturbation in the gene expression process primarily stems from DNA damage, as IdU is randomly integrated into the DNA chain, thereby inducing DNA damage (Li C et al., 2024 [DOI](#)). Although DNA damage caused by IdU treatment has little impact on genome-wide mean gene expression, it increases variance across the genome, thereby affecting cell differentiation (Desai RV et al., 2021 [DOI](#)). This is different from the understanding that increases in variance are usually caused by fluctuations in the mean gene expression (Newman JR et al., 2006 [DOI](#)). Meanwhile, the mean can be deconvolved as the product of BS and BF. This raises the question of the relationship between bursting dynamics and variance on a genome-wide scale, and how changes in bursting dynamics affect cell differentiation.

We used the DeepTX framework to infer a set of preprocessed scRNA-seq data (see Methods 2.1) to obtain underlying burst kinetics. The difference between the statistics and distribution obtained by inference and the statistics and distribution of sequencing data is minimal, ensuring the correctness of the inference results (Supplementary Figure S5a-l). Subsequently, we observed considerable variations in BS and BF across different genes (Figure 4A [DOI](#), Supplementary Figure S6d-f). Notably, genes with significantly increased BS exhibit corresponding decreases in BF, suggesting relative stability in the average gene expression level (Supplementary Figure S6d-f). Additionally, we observed a significant increase in transcriptional BS for most genes in the treatment group (Figure 4A [DOI](#)). This indicates that the rise in variance primarily results from the heightened BS induced by DNA damage (Figure 4B [DOI](#)). Notably, this conclusion is consistent with the results of theoretical analysis (Supplementary text 2).

To discover the biological characteristics of BS enhancement, we identified genes with upregulated BS using a *t*-test-based differential analysis method. Further, we perform GO enrichment analysis (Consortium GO, 2015 [DOI](#)) on the identified genes. We observed that the genes with enhanced BS were mainly enriched in the signal transduction of DNA damage response and the pathway that regulates the cell mitosis phase transition cycle (Smits VA & Medema RH, 2001 [DOI](#)) (Figure 4D [DOI](#)). Subsequently, we performed Gene Set Enrichment Analysis (GSEA) (Subramanian A et al., 2005 [DOI](#)) on the identified differentially expressed genes associated with BS and also observed enrichment in pathways related to cell cycle transitions and mitotic intra-S DNA damage checkpoint signaling (Figure 4E, F [DOI](#)). Similarly, we subjected genes significantly enhanced by BF in the treatment group (Supplementary Figure S6f) to GO analysis (Supplementary Figure S6g). Notably, the insignificance of the cell cycle transition pathway suggests that it is primarily influenced by BS-enhanced genes. Moreover, previous experimental studies have confirmed that treatment with IdU induces DNA damage, subsequently reprogramming cell differentiation fate (Rosina M et al.,

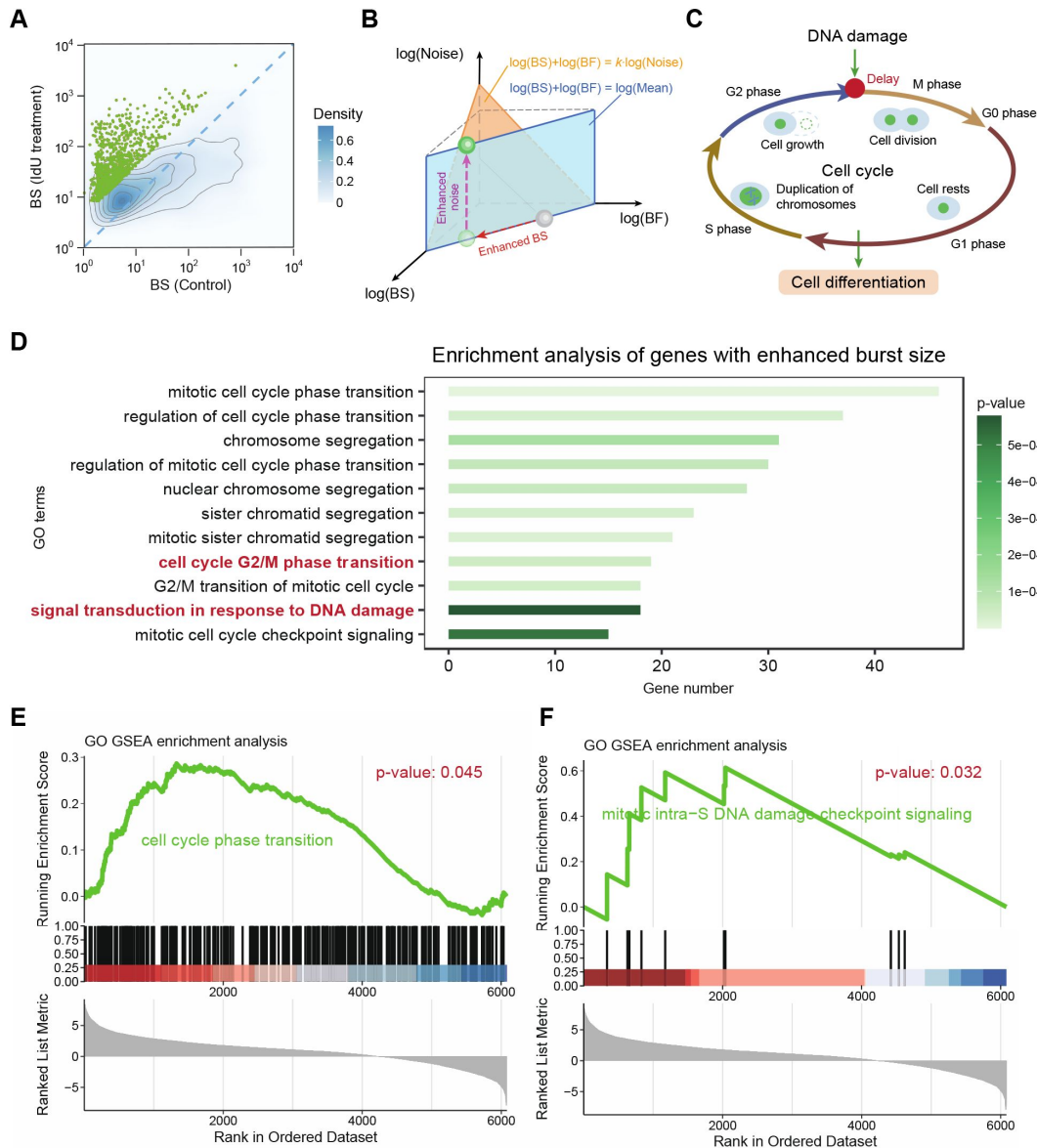


Figure 4.

Burst size enhancer delays cell cycle.

(A) Density map of BS of genes treated with IdU drugs compared to BS of genes not treated with drugs. Green dots represent up-regulated BS differential genes. **(B)** Schematic representation of the simultaneous enhancement of both BS and noise of a gene accompanied by the mean value of the gene remaining unchanged after IdU drug treatment. In the log-log-log 3D space, the mean expression level lies on a blue plane defined by $\log(\text{BS}) + \log(\text{BF}) = \log(\text{Mean})$, indicating that it is determined by the product of burst size and burst frequency. The orange plane captures a scaling relationship between expression noise and burst kinetics, described by $\log(\text{BS}) + \log(\text{BF}) = \kappa \log(\text{Noise})$, where κ is a constant reflecting their covariation. Notably, the trajectory of the green sphere shows that, under a fixed mean expression level (i.e., confined to the blue plane), increased gene expression noise primarily results from an increase in burst size. **(C)** Schematic diagram of the mechanism by which IdU drug treatment affects the mitotic cell cycle transition of cells via DNA damage. **(D)** Gene GO enrichment analysis is performed on the green dots of BS in (A) to obtain enrichment pathway diagram. The bold red pathways highlight the key components involved in the cellular differentiation mechanism. **(E, F)** Pathways obtained by GSEA enrichment analysis of BS of genes.

2019 [\[1\]](#); Riccio F et al., 2022 [\[2\]](#); Liu Z et al., 2017 [\[3\]](#)) (**Figure 4C** [\[4\]](#)). Additionally, research indicates that enhanced cell differentiation results from delayed cell cycle transitions (Rosina M et al., 2019 [\[5\]](#)).

DeepTX reveals that the BF enhancer induced by low-dose 5FU treatment accelerates cell apoptosis

The chemotherapeutic compound, 5-fluorouracil (5FU), exerts its cytotoxic effects on colorectal cancer cells by inflicting damage to their DNA (Kuipers EJ et al., 2015 [\[6\]](#); Bunz F et al., 1999 [\[7\]](#); Chang GS et al., 2014 [\[8\]](#)). This damage triggers apoptosis, a process of programmed cell death, thus inhibiting the growth and proliferation of malignant cells (Kuipers EJ et al., 2015 [\[6\]](#); Bunz F et al., 1999 [\[7\]](#); Chang GS et al., 2014 [\[8\]](#)). However, the impact of 5FU-induced DNA damage on the gene expression process and the underlying mechanism of cell apoptosis remains unclear. In this section, we employ the DeepTX to investigate the effect of 5FU-induced DNA damage on the gene expression process in colorectal cancer cells with scRNA-seq data (Park SR et al., 2020 [\[9\]](#)).

First, we inferred the burst kinetics from the scRNA-seq data of cells treated with low doses (10 μ m) of 5FU and controls. After preprocessing the data (see Methods 2.2), we inferred the scRNA-seq data to obtain a distribution that fits the scRNA-seq data (Supplementary Figure S7a-l), as well as the underlying burst kinetics (**Figure 5A, B** [\[10\]](#)).

Further, we found that the mean difference between the two groups of scRNA-seq data was not significant (p -value = 0.5 of t -test), but the variance and BS were significantly different (p -value < 0.05 of t -test **Figure 5A** [\[11\]](#) and Supplementary Figure S8a-c). Then, we performed a differential analysis of the BS and BF of gene expression between the control cells and the cells treated with 10 μ M 5FU, and then performed GO gene enrichment analysis on differential genes (**Figure 5C** [\[12\]](#) and Supplementary Figure S8d-f). We found that BS down-regulated differential genes were mainly enriched in apoptosis pathways (Hongmei Z, 2012 [\[13\]](#)), electron exchange pathways, and oxidation-related pathways (**Figure 5E** [\[14\]](#) and Supplementary Figure S8g). Specifically, the enrichment analysis indicated that although the burst size of genes in 5FU-treated cells was generally larger than that in the control group, the genes primarily influencing cell apoptosis exhibited a down-regulated burst size.

Our enrichment analysis results are consistent with existing experimental results (**Figure 5D** [\[15\]](#)). First, studies in recent years have found that many anticancer drugs trigger apoptosis and necrosis of cancer cells by activating the production of reactive oxygen species (ROS) (Sun Y & Rigas B, 2008 [\[16\]](#)). And experiments have shown that 5FU can induce ROS production in colorectal cancer cells (oxidation-related pathways) (Laha D et al., 2015 [\[17\]](#); Bwatanglang IB et al., 2016 [\[18\]](#); Chenna S et al., 2022 [\[19\]](#)). Meanwhile, mitochondria are the main source of ROS production, and defects in the mitochondrial electron transport chain will increase ROS production (electron exchange pathways) (Adam-Vizi V, 2005 [\[20\]](#)). As a result, excessive production of ROS can cause mitochondrial DNA damage and nuclear DNA damage (**Figure 5D** [\[21\]](#)). In particular, when mitochondrial DNA is damaged, mitochondrial damage repair will further increase the pressure of ROS (**Figure 5D right** [\[22\]](#)) (Shokolenko IN et al., 2014 [\[23\]](#)). Hence, DNA damage ultimately leads to cell death through apoptosis (apoptosis pathways) (**Figure 5D** [\[24\]](#)) (Handali S et al., 2018 [\[25\]](#); Hwang PM et al., 2001 [\[26\]](#)).

DeepTX reveals that BF enhancer induced by high-dose 5FU treatment contributes to cell survival

Low-dose 5FU treatment can induce cell apoptosis, while research shows high-dose 5FU treatment can induce cell resistance and evade apoptosis to survive (Kuranaga N et al., 2001 [\[27\]](#)). Hence, this part mainly studies the impact of differences in burst kinetics of gene expression between high-dose (50 μ m) 5FU-treated and control colorectal cancer cells on cell fate decisions.

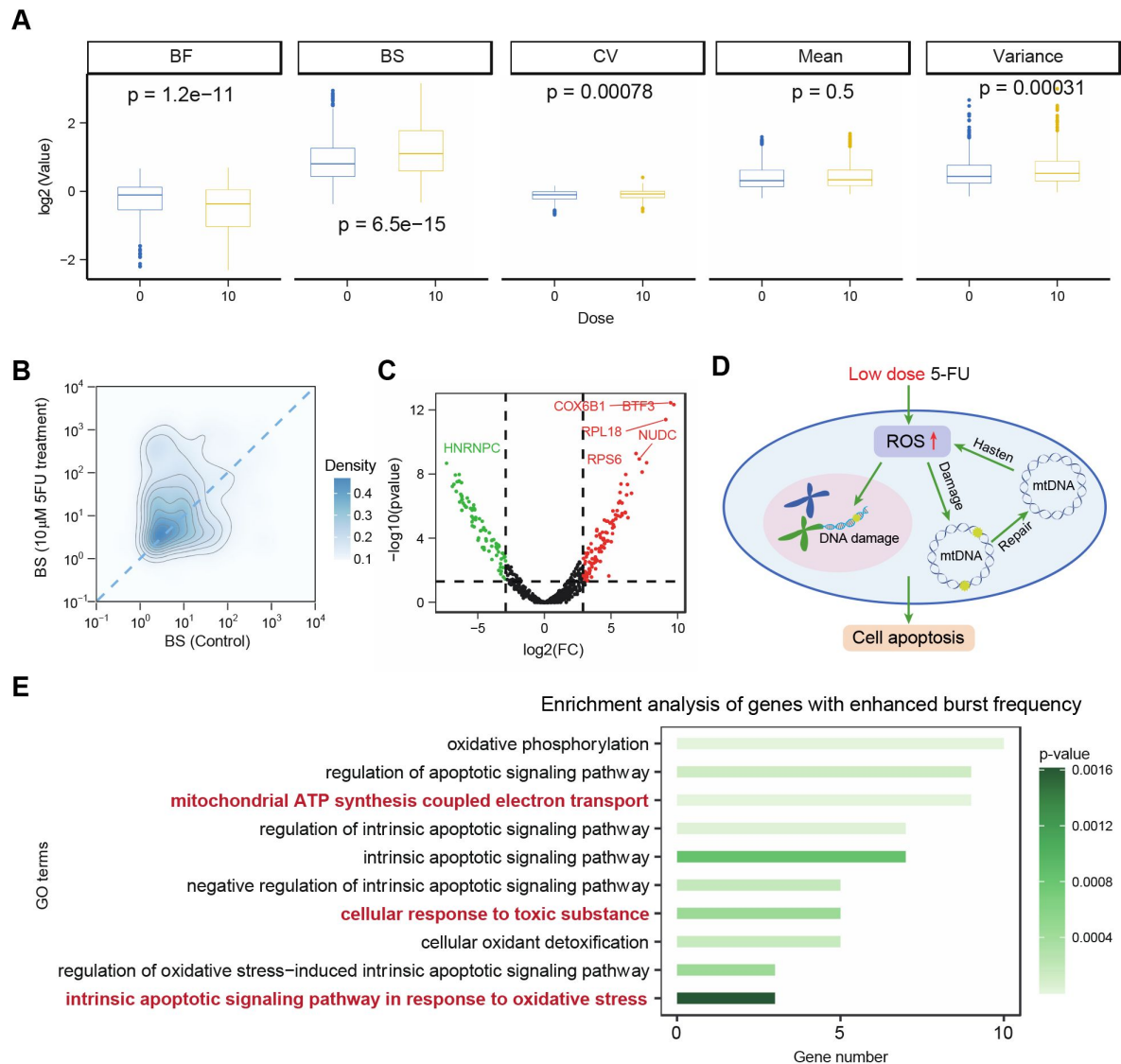


Figure 5.

Low dose 5FU treatment induces cell apoptosis by enhancing burst frequency.

(A) Comparison of gene bursting kinetics and statistics between control and 10-dose 5FU treated cells. (B) Density map of inferred burst size. (C) Volcano map of the fold changes (FC) of inferred burst size between control and 10-dose 5FU treated cells. (D) Mechanism diagram of apoptosis induced by low-dose 5FU drug treatment. (E) The enrichment analysis results were derived from genes that were downregulated in BS and upregulated in BF in the experimental group. The bold red pathways highlight the key components involved in the cellular apoptotic mechanism.

After data preprocessing (see Methods 2.2), we similarly observed the phenomenon that the average expression value of genes remained stable, while there is an enhanced increase in the variance genome-wide (Supplementary Figure S10a-c). Then, after preprocessing the data (see Methods 2.2), we inferred the scRNA-seq data to obtain the distribution that fits the scRNA-seq data (Supplementary Figure S9a-l), as well as the underlying burst kinetics (**Figure 6A**). Further, we performed a difference analysis on the BS of the two sets of inferred data, and selected genes with large differences in BS and BF for GO gene enrichment (**Figure 6A, B** and Supplementary Figure S10d-f). We found that the BS down-regulates differential genes were mainly enriched in the Cajal body related pathway, the electron exchange pathway, and telomerase-related pathways (Nabetani A & Ishikawa F, 2011), but the apoptosis-related pathways were not significant (**Figure 6D**). In particular, according to experimental studies, telomerase-related pathways can activate telomerase and play an important role in the elongation of telomeres (Cristofari G et al., 2007).

Gene enrichment results are consistent with existing ways of cells evading apoptosis (**Figure 6C**). That is, endogenous ROS production exceeds the cellular antioxidant defense capacity, leading to chemical damage to mitochondrial DNA (mtDNA). Meanwhile, high doses of 5FU causes Human Telomerase Reverse Transcriptase (hTERT) translocation, which on the one hand triggers telomerase activity and maintenance pathways, which enables cells to evade apoptosis (Kuranaga N et al., 2001). On the other hand, this translocation will enrich hTERT in mitochondrial DNA, thereby reducing oxidative stress and mitochondrial DNA damage to protect mitochondria and prevent cells from apoptosis (Lipinska N et al., 2017).

An important hallmark of cancer cell resistance is that cells are temporarily arrested in a non-proliferative G0 state (Wiecek AJ et al., 2023). To verify the presence of resistant cells in cells treated with high doses of 5FU, we selected genes related to G0 arrest and performed GSVA (Hänzelmann S et al., 2013) to obtain the G0 arrest quality score (Wiecek AJ et al., 2023) of each cell and perform UMAP dimensionality reduction visualization (**Figure 6E**). Through arrest quality score discrimination, we found that the proportion of resistant cells was large among cells treated with high doses of 5FU (**Figure 6F**). This result was consistent with the results of our enrichment analysis. Specifically, by comparing the BS and mean values of these genes, we found that most of the BS of G0 arrest related genes were up-regulated, but the mean changes were small (**Figure 6G**).

Discussion

Gene expression is inherently stochastic, leading to inter-individual variations primarily attributed to differences in burst patterns (Dar RD et al., 2012; Zenklusen D et al., 2008; Phillips R et al., 2012). Experimental observations confirm this bursting mechanism as a cellular response to environmental changes, particularly DNA damage (Friedrich D et al., 2019; Desai RV et al., 2021), influencing cell fate decisions. In this study, we introduced a mechanism-based deep learning framework (DeepTX) for performing genome-wide inference on transcriptional burst kinetics of a comprehensive gene expression model using scRNA-seq data. This framework offers an efficient computational approach to explore how cells regulate transcriptional burst kinetics in response to DNA damage at the genome-wide level, affecting various cell fate decisions.

Utilizing DeepTX, we analyzed three sets of scRNA-seq data comprising control and stimulus pairs driving different cell fate decisions: cell differentiation, apoptosis, and survival. This analysis unveiled significant insights into the underlying mechanisms of transcriptional burst kinetics. Firstly, IdU drug treatment-induced BS enhancement of genes may be associated with a delayed transition in the cell mitosis phase, which could potentially influence cell reprogramming and differentiation. Interestingly, while prior research suggested noise enhancement as the driver of cell fate decisions, we uncovered that BS enhancement is the deeper mechanism leading to cell differentiation, potentially resulting in cardiomyocyte differentiation. Secondly, low-dose 5FU

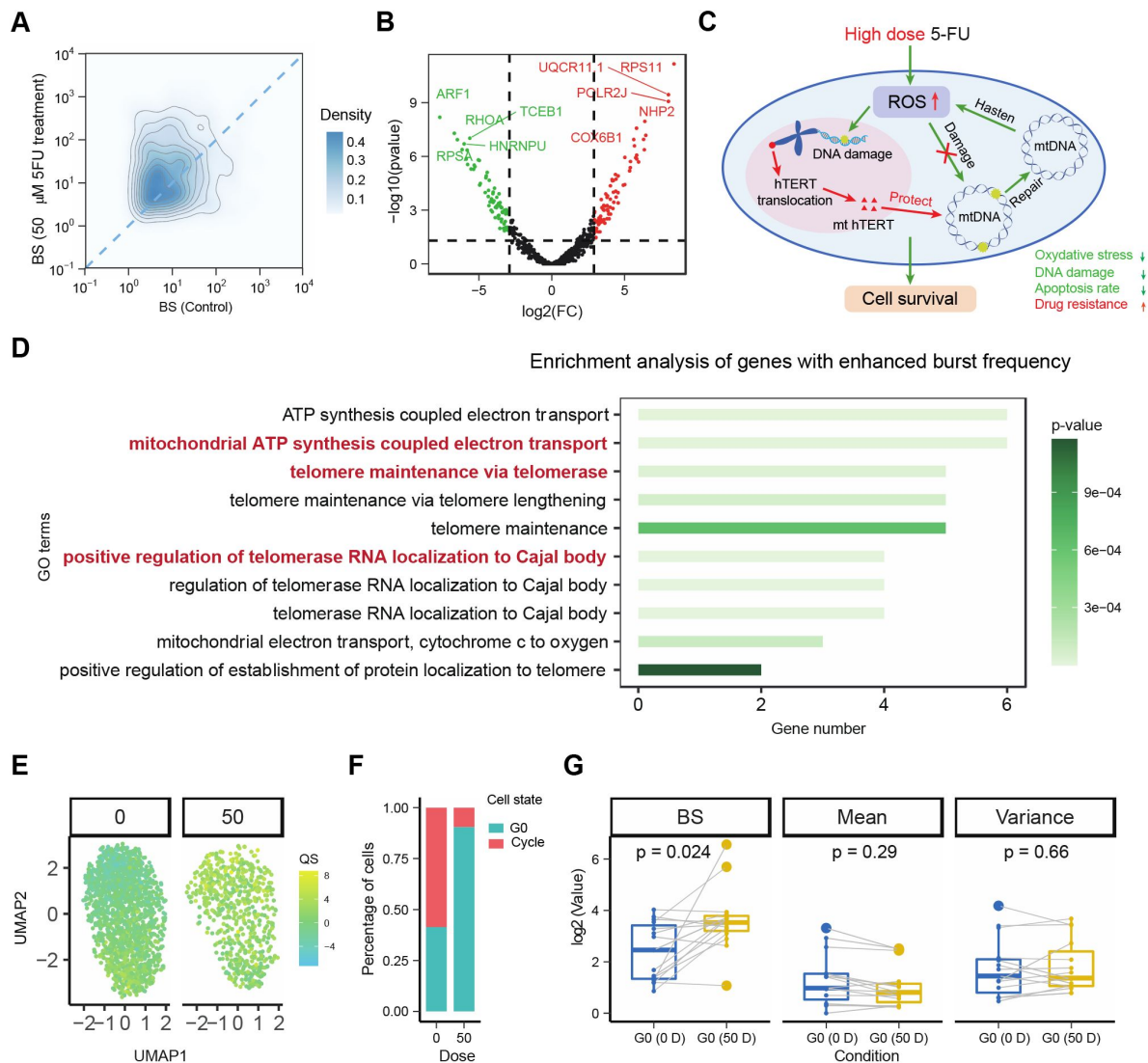


Figure 6.

High dose 5FU treatment may promotes cell drug resistance by altering burst frequency.

(A) Density map of inferred burst size. (B) Volcano map of the fold changes (FC) of inferred burst size between control and 50-dose 5FU treated cells. (C) Schematic diagram of the mechanism by which 5FU drug treatment induces cells to produce antioxidant properties and then continue to survive. (D) The enrichment analysis results were derived from genes that were downregulated in BS and upregulated in BF in the experimental group. (E) UMAP of G0 arrest quality score (QS) of control and 50-dose 5FU treated cells. (F) Proportion of G0 arrested (drug-resistant) and cycling (non-resistant) cells. (G) Comparison of gene bursting kinetics and statistics of G0 arrested genes between control and 50-dose 5FU treated cells. The light grey lines indicate the trend of changes in the corresponding bursting kinetics values, mean and variance for each gene.

treatment increases oxidative stress, promoting cell apoptosis by augmenting BF. Consistent with previous findings attributing apoptosis to ROS stress, our study reveals that ROS-induced DNA damage alters gene expression burst kinetics, ultimately triggering apoptosis. Thirdly, high-dose 5FU treatment induces telomerase extension, mitigating oxidative stress and fostering drug resistance by upregulating BF. Despite prior evidence of drug-resistant cells, our enrichment analysis underscores telomerase extension's role in resisting oxidative stress, underscoring the impact of BF enhancement.

We highlight several advantages of our DeepTX framework, which differs significantly from previous work that utilizes scRNA-seq data to study diverse noise sources from several aspects (Eling N et al., 2019 [DOI](#); Faure AJ et al., 2017 [DOI](#); Morgan MD & Marioni JC, 2018 [DOI](#); Ochiai H et al., 2020 [DOI](#)). Firstly, DeepTX employs a mechanistic hierarchical model that is both interpretable and extensible. To enhance the model's realism in interpreting scRNA-seq data, we developed a novel approach integrating the sequencing process and the underlying gene expression process. Specifically, we employed the TXmodel, a complex non-Markovian stochastic dynamic model that extends the waiting times for gene OFF and ON states into a non-exponential distribution, reflecting the multi-step nature of gene regulation. As a result, TXmodel not only captures gene expression dynamics under DNA damage but also depicts other multi-step regulatory processes, such as chromatin opening (Lomvardas S & Thanos D, 2002 [DOI](#)), preinitiation complex formation (Fuda NJ et al., 2009 [DOI](#)), and transcription factors binding (Nordick B et al., 2022 [DOI](#)). Secondly, despite the mathematical complexity of the mechanistic hierarchical model, DeepTX renders it tractable. This is achieved by (i) directly mapping model parameters to corresponding stationary distributions via a neural network trained on extensive simulated single-cell RNA data and theoretical analysis summary statistics for TXmodel; (ii) approximating the stationary distribution output of TXmodel, coupled with the sequencing process, by a mixed negative binomial distribution, maintaining the final distribution format. Thirdly, DeepTX's inference process is scalable, allowing genome-wide bursting kinetics inference from static scRNA-seq data. Despite the computational complexity of parameter inference involving hierarchical models and trained neural networks, DeepTX retains auto-differentiation for efficient and accurate gradient computation.

The DeepTX research paradigm is poised for extension to encompass both dynamic models and biological data. On the one hand, our focus is currently confined to the transcriptional processes influenced by DNA damage, limiting the explanatory scope of the TXmodel to larger-scale biological phenomena such as alternative splicing (Wan Y et al., 2021 [DOI](#); Gorin G & Pachter L, 2022 [DOI](#)), protein translation (Luo S et al., 2023 [DOI](#); Golan-Lavi R et al., 2017 [DOI](#)), epigenetic modification (Nicolas D et al., 2018 [DOI](#)), and chromatin movement (Liu T et al., 2016 [DOI](#); Wang Z et al., 2024 [DOI](#); Wang Z et al., 2023 [DOI](#)). Moreover, further development is needed to incorporate additional regulatory factors, including signaling pathways (Garrido-Rodriguez M et al., 2021 [DOI](#)), cell cycle progression, cell size (Sukys A & Grima R, 2025 [DOI](#)), and leaky transcription (Kepler TB & Elston TC, 2001 [DOI](#)). These broader biological processes necessitate more intricate dynamic models, often challenging to solve and thus challenging to apply to realistic biological data.

On the other hand, inference from static scRNA-seq relies on the steady-state assumption, which has some limitations. First, in some scenarios, the cell system may exhibit highly transient transcriptional programs that do not satisfy stationarity, leading to biased or misleading parameter estimates. For example, immediately following a synchronized developmental stimulus such as serum shock induced activation of immediate-early genes, transcription levels increase significantly (Hope B et al., 1992 [DOI](#)). Second, because DeepTX infers the mean burst frequency and size across the population, it cannot recover the underlying time-resolved dynamics or distinguish heterogeneous kinetic subpopulations. Third, lack of time-resolved measurements may affect the accuracy of inferences about dynamic parameters, especially the unidentifiability of parameters inferred from steady-state distributions, i.e., multiple parameters leading to the same steady-state distribution. The unidentifiability of individual parameters is a common and critical problem in

systems biology studies. Hence, in DeepTX, we employed a Bayesian approach based on loss potential to infer the posterior distributions of the parameters. Although DeepTX also encounters the issue of unidentifiability for individual parameters (Supplementary Figure S11), the multimodal nature of the posterior distribution suggests that multiple distinct parameter sets can produce similarly good fits to the observed data, highlighting the inherent non-identifiability of the model. Nevertheless, in the multimodal posterior distribution, at least one of the posterior peaks aligns closely with the ground truth, thereby demonstrating the validity of the inferred result. In future work, integrating time-series single-cell measurements with other emerging data types could help overcome these limitations.

Examples include increasingly comprehensive transcriptome data with both temporal and spatial resolution (Wen L & Tang F, 2022 [DOI](#)), as well as state-of-the-art multi-omics data (Miao Z et al., 2021 [DOI](#); Maizels RJ, 2024 [DOI](#)).

In conclusion, DeepTX provides a step toward inferring complex transcriptional kinetics from scRNA-seq data through mechanism-based deep learning inference. It sheds light on the augmented burst kinetics in DNA damage response across various cell fate decisions, thereby offering novel insights.

Methods

1. The DeepTX framework

1.1 Framework overview

The DeepTX Framework aims to infer the underlying bursting dynamics from static single-cell RNA sequencing (scRNA-seq) data to explore the impact of changes in bursting dynamics on cell fate decisions. The DeepTX framework includes two modules, DeepTXsolver and DeepTXinferer. DeepTXsolver can efficiently and accurately solve transcription models using neural network architecture, and DeepTXinferer can infer scRNA-seq data to obtain its underlying transcription burst kinetics using Bayesian methods.

The input of the DeepTXsolver is the parameters of the mechanism model, and the output is the corresponding stationary distribution solution of the model. The input to the DeepTXinferer is the discrete probability distribution of gene expression obtained from scRNA-seq data, and the output of the model is the parameters of the mechanism model.

The construction of the DeepTX framework consists of the following parts: (1) Model the gene expression process. (2) Train a neural network that solves the mechanism model using the data set generated by stochastic simulation algorithm (SSA) simulation. (3) Hierarchical model modeling to infer underlying burst kinetic parameters from scRNA-seq data.

When building the DeepTX Framework, we made the following assumptions: (1) The gene expression of cells was in a stationary distribution during sequencing. This assumption has been widely adopted in scRNA-seq studies, as it effectively captures mRNA expression distributions and enables the inference of underlying dynamic parameters. (Larsson AJM et al., 2019 [DOI](#); Luo S et al., 2023 [DOI](#); Ramsköld D et al., 2024 [DOI](#); Gupta A et al., 2022 [DOI](#)) (2) Gene expression counts of the same cell type follow the same distribution. Given that cells of the same type typically share similar transcriptional programs, their gene expression distributions are approximately identical, supporting the validity of this commonly used assumption. (3) The solution of the model can be approximated by a mixed negative binomial distribution. Since the stationary solution of chemical master equation can be expressed as a Poisson mixture, which can be more efficiently

approximated using negative binomial mixtures (Gorin G et al., 2024). (4) The state space sampled from a sufficiently long single simulation is statistically equivalent to that obtained from multiple simulations at steady state in gene expression models (Kuntz J et al., 2021).

1.2 Modelling stochastic gene expression processes

To recover dynamic transcriptional burst kinetic parameters from static scRNA-seq data, we require the stochastic modeling approach to profile the gene expression process. The traditional telegraph model is often used to model gene transcriptional bursting, which assumes that switching between genes follows an exponential distribution (i.e., switching is a single-step process) (Peccoud J & Ycart B, 1995; Kim JK & Marioni JC, 2013). However, this assumption is an oversimplification of the gene expression process in the context of DNA damage. Due to the fact that DNA damage slows down and even stops the elongation of RNA pol II across the double-strand DNA (Lans H et al., 2019), the gene expression process should be regarded as a multi-step process. This property causes the waiting time for the gene switching between active and inactive states to follow a non-exponential distribution (Daigle BJ, Jr. et al., 2015; Zhang J & Zhou T, 2019; Zhang JJ & Zhou TS, 2019). Therefore, we use TXmodel (Luo S et al., 2023) to better characterize the gene transcription process in the context of DNA damage (Figure 2), whose chemical reaction network (CRN) can be written as follows:



Where R_1 and R_2 describe the gene-state switching randomly between the ON state and OFF state, and $f_{\text{off}}(t, \omega_{\text{off}})$ and $f_{\text{on}}(t, \omega_{\text{on}})$ are the distribution of dwell time in ON and OFF state, respectively. Specifically, dwell time distribution refers to the probability distribution of the time in which a gene remains in a particular transcriptional state (ON or OFF) before transitioning to the other state. While the gene is in the active state, the synthesis of mRNA molecules is governed by a Poisson process with a constant synthesis rate k_{syn} (R_3). Meanwhile, each mRNA molecule can be stochastically degraded (R_4) with a constant degradation rate k_{deg} . Overall, the TXmodel can be determined by a set of parameters, denoted as $\theta = (\omega_{\text{off}}, \omega_{\text{on}}, k_{\text{syn}}, k_{\text{deg}})$.

The TXmodel as an extension of the traditional telegraph model can capture more realistic transcriptional burst kinetics, with BF and BS being two important parameters. The definition of BF is

$$\text{BF} = \frac{1}{\langle \tau_{\text{on}} \rangle + \langle \tau_{\text{off}} \rangle}, \quad (2)$$

Where $\langle \tau_{\text{off}} \rangle = \int_0^\infty t f_{\text{off}}(t, \omega_{\text{off}}) dt$ and $\langle \tau_{\text{on}} \rangle = \int_0^\infty t f_{\text{on}}(t, \omega_{\text{on}}) dt$ are the mean dwell time in OFF and ON state, respectively. BF represents the average time it takes for the promoter to complete one full stochastic cycle between its active and inactive states. This definition is slightly different from the traditional definition $1/(k_{\text{deg}} * \tau_{\text{off}})$, which can be regarded as a simplified version of our definition (Eq. 2), under the assumptions that τ_{on} is negligible and

$k_{\text{deg}} = 1$ (i.e., rate parameters are normalized to be dimensionless). Although it is reasonable to neglect activation time τ_{on} , as it is typically much shorter than inactive time under some conditions, we chose a more complete way to define the burst frequency so that it is applicable to more general situations. Additionally, Eq. 2 has been widely used in recent literature (Ramsköld D et al., 2024; Zoller B et al., 2018; Hoppe C et al., 2020). The definition of BS is

$$\text{BS} = k_{\text{syn}} \langle \tau_{\text{on}} \rangle. \quad (3)$$

BS refers to the average number of mRNA transcripts produced during a single transcriptional activation event of a gene. Notably, the mean transcription level of mRNA (y) can be expressed as $\langle y \rangle = k_{\text{syn}} \langle \tau_{\text{on}} \rangle (\langle \tau_{\text{on}} \rangle + \tau_{\text{off}}) = \text{BS} \times \text{BF}$.

Without loss of generality, the gamma distribution can be used to characterize non-exponential distributions, i.e., $f_{\text{off}}(t, \omega_{\text{off}}) = r_{\text{off}}^{k_{\text{off}}} t^{k_{\text{off}}-1} e^{-r_{\text{off}} t} / \Gamma(k_{\text{off}})$ such that $\omega_{\text{off}} = (r_{\text{off}}, k_{\text{off}})$ and $f_{\text{on}}(t, \omega_{\text{on}}) = r_{\text{on}}^{k_{\text{on}}} t^{k_{\text{on}}-1} e^{-r_{\text{on}} t} / \Gamma(k_{\text{on}})$ such that $\omega_{\text{on}} = (r_{\text{on}}, k_{\text{on}})$, where $\Gamma(\cdot)$ is Gamma function, r_{off} and r_{on} are the rate-parameters of state switching, k_{off} and k_{on} are the possible number of reaction steps between OFF and ON state. The rationale for employing the gamma distribution lies in the complex, multi-step nature of promoter state transitions. Biologically, both the “ON” and “OFF” states of a promoter encompass multiple underlying sub-states, with transitions governed by a series of molecular events, each characterized by exponentially distributed waiting times. When aggregated, these sequential exponential processes yield non-exponential overall waiting time distributions. The gamma distribution, being a natural analytical extension of such a convolution of multiple exponential distributions, provides a mathematically and biologically appropriate approximation for such processes. Here, in the TX model, the state-switching waiting time follows a non-Markovian distribution (i.e., it is not exponentially distributed), indicating that the system exhibits non-Markovian dynamics.

It is a challenge to solve the stationary distribution of this gene expression system directly from a theoretical approach. But we proved that the binomial moments of this system can be obtained in an iterative form, by rationalizing this gene expression system as a queueing theory model. Thus, we can theoretically compute some important summary statistics $\mathbf{s}(\boldsymbol{\theta})$ given a parameter $\boldsymbol{\theta}$, such as mean, variance, kurtosis, and bimodal coefficient (see details in Supplementary Text 1.2).

1.3 Training neural network to solve gene expression model

The stationary distribution of CRN (1) is difficult to be solved analytically. In general, stochastic simulation algorithms are used to approximate the solution of chemical master equations (Boguñá M et al., 2014 [\[14\]](#); Masuda N & Rocha LE, 2018 [\[15\]](#)). However, the inference process involves updating the parameters, and each update requires a simulation to obtain the stationary distribution, which makes statistical inference time-consuming. We note that deep learning methods may have potential applications in solving this challenge because it is widely used for approximating the solution of partial differential equations, chemical master equations, and non-Markovian chemical master equations (Jiang Q et al., 2021 [\[16\]](#); Wang S et al., 2019 [\[17\]](#); Davis CN et al., 2020 [\[18\]](#); Gupta A et al., 2021 [\[19\]](#); Bar-Sinai Y et al., 2019 [\[20\]](#)).

Therefore, in this part, we used a neural network to approximately solve CRN (1). This enables CRN (1) to be solved efficiently and accurately. Details will be introduced in the following part.

Generating synthetic training dataset

To construct the mapping from the parameters of TXmodel to the stationary distribution, we first need to generate the neural network's training set, i.e., a large-scale range of parameter sets as features and corresponding probability distributions as labels. For the features, we used the Sobol algorithm (Sobol IM, 1967 [\[21\]](#)) to conduct large-scale sampling to generate the neural network data set, which can cover the entire parameter space more evenly than random sampling.

For the labels, we perform the SSA of the TXmodel to obtain the stationary distribution given a set of preset parameters. For clarity, the synthetic training dataset is denoted by $\{(\boldsymbol{\theta}_i, P_{\text{simulation},i})\}$, where $\boldsymbol{\theta}_i$ presents the model parameters of TXmodel, and $P_{\text{simulation},i}$ represents the stationary distribution solution obtained by SSA (Figure 2A [\[22\]](#)). Specifically, the value range of each parameter component of $\boldsymbol{\theta}_i$ is $k_{\text{off}} \in (1, 15)$, $r_{\text{off}} \in (0.1, 10)$, $k_{\text{on}} \in (1, 15)$, $r_{\text{on}} \in (0.01, 10)$, $k_{\text{syn}} \in (0.1, 400)$. The experimentally measured values of the mean duration of the OFF state, burst duration,

transcriptional burst sizes, and the number of intermediate steps in transcriptional state transitions fall within the specified parameter ranges (Tunnacliffe E & Chubb JR, 2020 [DOI](#); Gupta A et al., 2022 [DOI](#)).

Deep learning model architecture

We build the basic deep learning model architecture in DeepTX via fully connected neural networks, which has been effectively used to solve approximate solutions from biochemical reaction networks (Sukys A et al., 2022 [DOI](#); Gupta A et al., 2021 [DOI](#)). The deep learning model architecture consists of an input layer, hidden layers, and an output layer.

Firstly, the input layer contains 5 neurons corresponding to the parameters of the mechanism model $\theta_i = (k_{\text{off},i}, r_{\text{off},i}, k_{\text{on},i}, r_{\text{on},i}, k_{\text{syn},i})$. Secondly, the input layer is followed by n_{hidden} hidden layers with each layer containing $n_{\text{neurons}} = (n_{\text{neurons},1}, \dots, n_{\text{neurons},n_{\text{hidden}}})$ neurons. For the neurons in each hidden layer, we implement a nonlinear mapping by the commonly used ReLU activation function (Supplementary Figure S1):

$$\begin{aligned} h_{ij}^{(1)} &= \text{ReLU}(w_{1j}^{(1)}k_{\text{off},i} + w_{2j}^{(1)}r_{\text{off},i} + w_{3j}^{(1)}k_{\text{on},i} + w_{4j}^{(1)}r_{\text{on},i} + w_{5j}^{(1)}k_{\text{syn},i}), (j=1,2,\dots,n_{\text{neurons},1}), \\ h_{ij}^{(2)} &= \text{ReLU}(w_{1j}^{(2)}h_{i1}^{(1)} + w_{2j}^{(2)}h_{i2}^{(1)} + \dots + w_{n_{\text{neurons},1,j}}^{(2)}h_{i,n_{\text{neurons},1}}^{(1)}), (j=1,2,\dots,n_{\text{neurons},2}), \\ &\vdots \\ h_{ij}^{(n_{\text{hidden}})} &= \text{ReLU}(w_{1j}^{(n_{\text{hidden}})}h_{i1}^{(n_{\text{hidden}}-1)} + w_{2j}^{(n_{\text{hidden}})}h_{i2}^{(n_{\text{hidden}}-1)} + \dots + w_{n_{\text{neurons},n_{\text{hidden}}-1,j}}^{(n_{\text{hidden}})}h_{i,n_{\text{neurons},n_{\text{hidden}}-1}}^{(n_{\text{hidden}}-1)}), (j=1,2,\dots,n_{\text{neurons},n_{\text{hidden}}}), \end{aligned} \quad (4)$$

Where $h_{ij}^{(n_{\text{hidden}})}$ represents the value of the j -th neuron of the n_{hidden} -th hidden layer of the i -th group of parameters, $w_{kj}^{(n_{\text{hidden}})}$ represents the neural network weight connecting the k -th neuron of the n_{hidden} -th hidden layer and the j -th neuron of the $(n_{\text{hidden}}-1)$ -th hidden layer.

Thirdly, In the output layer, we aim to allow neurons to approximate synthetic scRNA-seq data generated by TXmodel. Negative binomial distributions are a class of distributions that are widely used in modeling scRNA-seq data and have been shown to account for both the “drop-out” phenomenon and the fact that the Fano factor is greater than 1 in scRNA-seq data (Risso D et al., 2018 [DOI](#); Love MI et al., 2014 [DOI](#)). Also, the negative binomial distribution has been employed as an approximate stationary solution to the TXmodel for many gene expression systems (Sukys A et al., 2022 [DOI](#); Öcal K et al., 2022 [DOI](#)). We assign weights and parameters (a_{ij} , m_{ij} , l_{ij}) to the output layer neurons (Supplementary Figure S1):

$$\begin{aligned} h_{ij}^{(\text{output})} &= w_{1j}^{(\text{output})}h_{i1}^{(n_{\text{hidden}})} + w_{2j}^{(\text{output})}h_{i2}^{(n_{\text{hidden}})} + \dots + w_{n_{\text{neurons},n_{\text{hidden}}},j}^{(\text{output})}h_{i,n_{\text{neurons},n_{\text{hidden}}}}^{(n_{\text{hidden}})}, (j=1,2,\dots,n_{\text{output}}), \\ (a_{i1}, a_{i2}, \dots, a_{i,n_{\text{NB}}}) &= \text{Softmax}(h_{i1}^{(\text{output})}, \dots, h_{i,3*j+1}^{(\text{output})}, \dots, h_{i,3*n_{\text{NB}}-2}^{(\text{output})}), \\ m_{ij} &= \text{Exp}(h_{i,3*n_{\text{NB}}-1}^{(\text{output})}), j=1,\dots,n_{\text{NB}}, \\ l_{ij} &= \text{Sigmoid}(h_{i,3*n_{\text{NB}}}^{(\text{output})}), j=1,\dots,n_{\text{NB}}, \end{aligned} \quad (5)$$

Where $h_{ij}^{(\text{output})}$ represents the value of the j -th neuron of the output layer of the i -th set of parameters, a_{ij} is the weight coefficient of the negative binomial distribution, satisfying $\sum_{i=1}^{n_{\text{NB}}} a_{i1} = 1$, l_{ij} , m_{ij} represents the i -th parameter of the j -th negative binomial distribution of the mixed negative binomial distribution, n_{NB} is the number of negative binomial distribution. And the mixed negative binomial distributions can be expressed as (Supplementary Figure S1):

$$P_{\text{neuralnet},i}(n_{\text{counts}}; \mathbf{w}(\theta_i)) = \sum_{j=1}^{n_{\text{NB}}} a_{ij} (\mathbf{w}(\theta_i)) \text{NB}(n_{\text{counts}}; m_{ij}(\mathbf{w}(\theta_i)), l_{ij}(\mathbf{w}(\theta_i))), \quad (6)$$

$$\text{where } \text{NB}(N = n_{\text{counts}}; l, m) = \binom{n_{\text{counts}} + m - 1}{n_{\text{counts}}} l^{n_{\text{counts}}} (1-l)^m.$$

Loss function of DeepTXsolver

Our loss function includes two parts: one part is the KL divergence between the mixed negative binomial distribution generated by neural network $P_{\text{neuralnet},i}$ and label distribution $P_{\text{simulation},i}$ generated by SSA, and the another part is the log difference of the moment statistics between theoretical solution of the TXmodel s_{ij} and the output of the neural network $\hat{s}_{ij}$:

$$\text{Loss} = \sum_{i=1}^{n_{\text{batch}}} \left(\text{KL}(P_{\text{simulation},i}, P_{\text{neuralnet},i}) + \lambda \sum_{j=1}^{n_{\text{stats}}} \left(\log \left(\frac{s_{ij}}{\hat{s}_{ij}} \right) \right)^2 \right), \quad (7)$$

Where $\text{KL}(P_{\text{simulation},i}, P_{\text{neuralnet},i}) = \sum_{n_{\text{counts}}} P_{\text{simulation},i} \log(P_{\text{neuralnet},i} / P_{\text{simulation},i})$, and λ is the hyperparameter, represents the weight coefficient of the statistical loss, and $\lambda = 0.1$, n_{batch} represents the number of samples, where each sample is characterized by $(\theta_i, P_{\text{simulation},i})$ in each batch, and n_{stats} is the number of summary statistics. For the latter part of loss function, we can still obtain the moment statistic by theoretical derivation, although the analytic distribution is difficult to solve. Adding moment statistics to the loss function can make the neural network more robust (Figure 2E). Specifically, we use four statistics here: s_{i1} indicates the mean, s_{i2} denotes the Fano factor, s_{i3} denotes kurtosis, s_{i4} represents the bimodal coefficient. Specifically, the Fano factor, which normalizes variance by the mean, provides a robust measure of transcriptional noise across genes or conditions with varying expression levels. It's worth mentioning that the statistics here is scalable and we can add more information inside the loss function.

Training details of DeepTXsolver

Effective model training necessitates the selection of an appropriate optimizer to minimize the loss function. In our case, we employed the Adam optimizer (Kingma DP & Ba J, 2014), a widely adopted choice in deep learning due to its robust performance. And each time, a small batch of samples is selected to update the neural network parameters $\mathbf{w}$ (Goodfellow I et al., 2016), and the iteration of all samples is called an epoch. Furthermore, we utilized the Glorot Uniform method to establish an optimal initial point for our neural network parameter (Glorot X & Bengio Y, 2010).

Batch size and learning rate are key optimization hyperparameters affecting model convergence. A smaller batch size enhances model generalizability but lengthens training time, hence a batch size of 64 is chosen for balance (Keskar NS et al., 2016). In gradient descent process, the learning rate significantly influences the step size, where large rates may induce instability or divergence, and conversely, small rates may precipitate premature convergence to local optima. To counteract this, a decaying learning rate is used, allowing larger early updates for faster convergence and preventing instability in later stages. Further, training of the model stops if it has been trained for over 200 epochs or if the learning rate has been reduced five times. These criteria are adaptive and efficient as further training does not significantly enhance the model's performance.

Hyperparameter tuning of DeepTXsolver

To obtain a neuron network model with accurate prediction and generalization, we compared the model architectures in terms of the number of neurons per layer, the number of hidden layers, and the number of neurons in the output layer, and compared the performance of the model with different dataset sizes. And we applied the Hellinger distance between the true distribution and the predicted distribution by the neuron network as a judgment criterion.

First, we compared the performance of architectures with a single hidden layer with (16, 32, 64, 128, 256, 512) neurons, and found that the performance of the model barely increased after increasing to 128 neurons (Supplementary Figure S2a). Secondly, after comparing the performance of architectures with different numbers of hidden layers, we found that a single hidden layer has the best performance (Supplementary Figure S2b). Thirdly, comparing the performance of $3i$ ($i = 1, 2, \dots, 10$) neurons in the output layer, we found that after 12 neurons (4 negative binomial distribution mixtures), the performance of the model was not improved (Supplementary Figure S2c). Particularly, four negative binomial distribution mixtures are able to fit the bimodal nature of the distribution as well as ensure good training efficiency (Öcal K et al., 2022 [DOI](#)).

Finally, to obtain a dataset that contributes to convergence as well as model generalization while ensuring training efficiency, we compared the model performance for dataset sizes of (500, 1000, 5000, 10000, 15000) and found that increasing the dataset size further does not significantly improve the model performance after to 15000 samples (Supplementary Figure S2d). Therefore, we identified a single hidden layer of 128 neurons and an output layer of 12 neurons as the appropriate model architecture, and 15,000 samples as the appropriate training set.

1.4 Inferring burst kinetics from scRNA-seq data with DeepTXinferrer

This section describes how to recover dynamic burst kinetics from static scRNA-seq data. In principle, once we have constructed the mapping of the parameters to the corresponding stationary distributions through the pre-trained neural network in [section 1.3](#) [DOI](#), we can quickly access the corresponding likelihood functions when updating the parameters during the statistical inference process. However, true scRNA-seq data is the result of a combination of gene expression processes and sequencing processes, and the current pre-trained neural network can only yield a description of the gene expression process modeled by TXmodel. Therefore, we require to extent it into a hierarchical model to depict the more realistic generation mechanism behind the scRNA-seq data (Sarkar A & Stephens M, 2021 [DOI](#)).

Modeling observed scRNA-seq data with hierarchical model

The hierarchical model aims to couple two different processes, specifically, the true gene expression mechanism process and the sequencing process that immediately follows. In terms of a formula, the probability distribution of observations can be represented as a convolution:

$$P(Y = y) = \int_0^\infty P_{\text{measure}}(y|x) P_{\text{model}}(x) dx, \quad (8)$$

where P_{measure} is the measure model and P_{model} is the expression model.

First, we model the sequencing process. Each cell can be regarded as a pool of mRNA, and $X_{cg}^{(\text{true})}$ represents the true expression amount of the g -th gene in the c -th cell. $Y_{cg}^{(\text{observed})}$ refers to the observed value obtained by scRNA-seq from the given true expression value $X_{cg}^{(\text{true})}$, satisfying the following conditional probability distribution

$$Y_{cg}^{(\text{observed})} | X_{cg}^{(\text{true})} \sim P_{\text{measure}}(y | x), \quad (9)$$

Where $P_{\text{measure}}(y | x)$ characterizes all noise in the sequencing process, and is generally assumed to be Binomial distribution or Poisson distribution, which has been proven experimentally and theoretically (Sarkar A & Stephens M, 2021 [\[1\]](#); Wang J et al., 2018 [\[2\]](#)). By incorporating an additional sampling probability, denoted as α_{cg} , into the sequencing process, we are able to characterize capture efficiency. We make the assumption that intercellular molecules are independent of one another, and that only proportional products are captured and sequenced, following a Binomial distribution

$$Y_{cg}^{(\text{observed})} | X_{cg}^{(\text{true})} \sim P_{\text{measure}}(y | x, \alpha_{cg}), \quad (10)$$

Where α_{cg} is the sampling probability. Although the experimental capture efficiency ranges from 0.06 to 0.32 (Zheng GX et al., 2017 [\[3\]](#); Macosko EZ et al., 2015 [\[4\]](#)), we fixed the parameter $\alpha_{cg} = 0.5$ to minimize complexity and unidentifiability issues. Further, we demonstrated that inference across different efficiencies (0.5, 0.3, 0.2) consistently yielded strong correlations between inferred burst size and burst frequency (Supplementary Figure S12).

Second, we model the true gene expression process, assuming that it satisfies the following probability distribution

$$X_{cg}^{(\text{true})} \sim P_{\text{model}}(x). \quad (11)$$

The selection of the true gene expression probability distribution is very critical. It needs to meet the following conditions: (i) It can well characterize the transcription burst kinetics. (ii) The inference derived from this distribution necessitates both efficiency and precision.

In [section 1.2](#) [\[5\]](#), we have constructed a model of the gene expression process with DNA damage, enabling the model's stationary distribution to encapsulate the bursty dynamics inherent in gene expression. The solution to the model is approximated by a mixed negative binomial distribution, which is obtained from a trained neural network. Importantly, this mixed negative binomial distribution enables efficient inference. Hence,

$$P_{\text{model}}(x) = P_{\text{neuralnet}}(x, \theta). \quad (12)$$

Further, the following hierarchical model can be obtained

$$\begin{aligned} Y_{cg}^{(\text{observed})} | X_{cg}^{(\text{true})} &\sim \text{Binomial}(y_{cg} | x_{cg}, \alpha_{cg}), \\ X_{cg}^{(\text{true})} &\sim \sum_{j=1}^{n_{\text{NB}}} a_j \text{NB}_j(x_{cg}, m_j(\theta), l_j(\theta)), \end{aligned} \quad (13)$$

Where $X_{cg}^{(true)}$ represents the random variable of the true expression value, $Y_{cg}^{(observed)}$ represents the random variable of the observed expression value, Binomial represents the binomial distribution, NB represents negative binomial distribution and α_{cg} represents the sampling probability. Further, we can obtain

$$Y_{cg}^{(observed)} \sim \sum_{n_{true}=1}^{n_{count}} \left(\text{Binomial}(y_{cg} | x_{cg}, \alpha_{cg}) \sum_{j=1}^{n_{NB}} a_j \text{NB}_j(x_{cg}, m_j(\theta), l_j(\theta)) \right). \quad (14)$$

It is proved that $Y_{cg}^{(observed)}$ also follows a mixed negative binomial distribution (Supplementary text 1.3), denoted as $P_{obsmodel}$.

Estimation and optimization process of DeepTXinferer

We can use the DeepTX framework to infer scRNA-seq data to obtain the corresponding gene expression burst kinetics through optimization methods. Given the expression counts of gene g in all cells, its corresponding probability distribution $P_{obsdata}$ can be obtained. We use the Hellinger distance to compare the error between the true probability distribution $P_{obsdata}$ of genes and the model probability distribution $P_{modeldata}$. The following optimization problem can be obtained

$$\hat{\theta} = \arg \min_{\theta} \sum_{m=0}^{n_{count}} \left(\sqrt{P_{obsmodel}(y_{cg} = m; \theta)} - \sqrt{P_{obsdata}(x = m)} \right)^2 \quad (15)$$

Where n_{count} is the maximum value of mRNA expression, $P_{obsdata}$ is the density probability for each gene g . Specifically, $P_{obsdata}$ is a function that is differentiable with respect to parameter θ_g . We can use the gradient descent method to obtain the optimal parameters of the TXmodel and the corresponding distribution of the TXmodel.

Posterior distribution

The optimization method in the previous section can obtain the parameters of the mechanism model that satisfies the distribution of scRNA-seq data. In this section, we give a method to solve the confidence interval of each optimized parameter (Gaskin T et al., 2023 [23](#)).

Each iteration of the optimization process can obtain the estimated value $\hat{\theta}^{(k)} (k = 1, \dots, n_{iteration})$ and loss value $J(\hat{\theta})$ corresponding to the estimated value. In particular, according to Equation (15), the loss is calculated as follows.

$$J(\hat{\theta}^{(k)}, \theta) = J^{(k)}(\hat{P}_{obsmodel}(\hat{\theta}^{(k)}), P_{obsdata}) = \sum_{m=0}^{n_{count}} \left(\sqrt{\hat{P}_{obsmodel}(x = m; \hat{\theta}^{(k)})} - \sqrt{P_{obsdata}(x = m; \theta)} \right)^2, \quad (16)$$

The following posterior probability can be obtained

$$\pi(\hat{\theta}^{(k)} | P_{obsdata}) \sim \frac{n^{(k)} \exp(-J^{(k)}(\hat{P}_{obsmodel}, P_{obsdata}))}{\sum_{k=1}^{n_{iteration}} \exp(-J^{(k)}(\hat{P}_{obsmodel}, P_{obsdata}))} \pi^{(0)}(\hat{\theta}^{(k)}), \quad (17)$$

where $\pi^{(0)}$ is the prior, we take it as uniform density, $n^{(k)}$ represents the number of values $\hat{\theta}^{(k)}$ takes during the iteration process, and $\exp(-J(\hat{P}_{mixture}, P_{simulation}))$ represents the loss potential of each iteration. Let the posterior probability be $P_{posterior}(\hat{\theta}^{(k)})$, and the marginal distribution corresponding to each component of parameter $\hat{\theta}^{(k)}$ can be obtained

$$p(\theta_j^{(k)}) \sim \int P_{posterior}(\hat{\theta}^{(k)}) d\hat{\theta}_{-j}^{(k)}. \quad (18)$$

The subscript $-j$ means that we integrate the components of $\theta^{(k)}$ except $\theta_j^{(k)}$. In particular, to allow the iterative parameters to traverse the entire parameter space, multiple optimizations can be performed with different initial values to improve the performance of the posterior distribution.

Validation on synthetic data

We conducted an evaluation of the inference performance of the DeepTX framework on synthetic scRNA-seq data, focusing on three key aspects. (i) Model accuracy: This refers to the minimal discrepancy between the distribution corresponding to the inferred parameters and the input distribution. (ii) Parameter identifiability: This denotes the proximity of the values of the inferred parameters to the true parameters, suggesting that the model can accurately identify the true parameters. (iii) Result robustness: This implies that the outcomes of each inference do not exhibit significant variations.

We created a synthetic dataset $\{(\theta_i^{\text{syn}}, P_i^{\text{syn}})\} (i = 1, 2, \dots, 500)$ using Sobol sampling and SSA under preset parameters. The synthesized distribution P_i^{syn} is used as input to the DeepTX framework, and upon optimization, we obtained estimated parameters and their corresponding distribution $\{(\theta_i^{\text{est}}, P_i^{\text{est}})\}$. We showed that the Hellinger distance between the estimated and synthetic dataset's distribution was minimal, indicating accurate estimations (Supplementary Figure S3a). Additionally, the burst kinetics of both the synthetic data and the estimated data can be computed utilizing their respective parameters, and θ_i^{est} . We showed that a strong correlation was observed between the synthetic burst kinetics $\gamma^{\text{syn}} = (\text{BF}^{\text{syn}}, \text{BS}^{\text{syn}})$ and estimated burst kinetics $\gamma^{\text{est}} = (\text{BF}^{\text{est}}, \text{BS}^{\text{est}})$ (Supplementary Figure S3b-c), suggesting that burst kinetics are identifiable.

Subsequently, we conducted a robustness verification of the framework's ability to infer burst kinetics. The error was computed as follows:

$$\text{Error}(\gamma^{\text{syn}}, \gamma^{\text{est}}) = \left(\log(\text{BF}^{\text{syn}}) - \log(\text{BF}^{\text{est}}) \right)^2 + \left(\log(\text{BS}^{\text{syn}}) - \log(\text{BS}^{\text{est}}) \right)^2. \quad (19)$$

Given a set of parameters θ_i^{syn} , we drew samples of sizes (100, 500, 1000, 1500, 2000) from the probability distribution P_i^{syn} corresponding to θ_i^{syn} . For each sample size, we performed 50 repetitions, and then DeepTX framework is used to infer the distribution obtained from each sampling, setting different initial values each time. We found that the inferred results are robust, and the robustness increases with the increase in the number of samples (Supplementary Figure S3d-e).

2 Dataset

2.1 IdU and DMSO dataset

To investigate the effects of 5'-iodo-2'-deoxyuridine (IdU) on genome-wide gene expression, Desai et al. conducted scRNA-seq data on mouse embryonic stem cells, both with and without IdU treatment. Consequently, they obtained scRNA-seq data for 12481 genes of transcriptomes from 812 cells and 13780 genes of transcriptomes from 744 cells, respectively. The authors found a phenomenon that IdU drug treatment increased transcriptional noise genome-wide but did not affect transcriptional mean, further affecting cell fate decisions.

We used the DeepTX framework to infer this set of scRNA-seq data to obtain potential transcription burst kinetics, and further study the impact of changes in transcription burst kinetics on cell fate decisions. To eliminate the impact of technical noise on data inference, we preprocess the data as follows. Firstly, We normalized the data y_{cg} using Seurat, where y_{cg} represents the expression level of the gene g in the cell c . The normalization process is that we multiplied each y_{cg} by a scaling factor $S = y_{\text{normalized}} / \sum_{g=1}^{n_{\text{genes}}} y_{cg}$, where $y_{\text{normalized}} = 50000$. Particularly,

the scaling factor S is related to the total expression of genes in each cell. Secondly, after normalizing the expression, we divided it into bins with a unit of 1 to obtain the bin where each expression value is located. Finally, to eliminate the impact of low-expression genes on the inference process, we filter out genes whose average gene expression is less than 1. After data processing, there are 6082 genes from 812 cells left in our IdU and 6082 genes from 744 cells left in DMSO data sets.

2.2 5FU Dataset

To study the impact of 5-fluorouracil (5FU) drug-induced DNA damage on cell fate decisions, Park SR et al. performed scRNA-seq on colon cancer cells treated with 0 μm , 10 μm , and 50 μm 5FU drug (Park SR et al., 2020 [DOI](#)) and obtained 8534 genes from 1673 cells, 7077 genes from 632 cells, and 6661 genes from 619 cells, respectively. The authors found that Colon cancer cells treated with different doses of 5FU exhibit different transcriptional phenotypes, and these different phenotypes correspond to different cell fate decisions.

We used the DeepTX framework to infer these three sets of scRNA-seq data to obtain potential transcriptional burst kinetics, thereby studying the impact of changes in transcriptional burst kinetics caused by different doses of 5FU treatment on cell fate decisions. To eliminate the impact of technical noise on data inference, we preprocess the data in the following steps.

First, we normalize y_{cg} by multiplying a scale factor S , where $S = y_{\text{normalized}} / \sum_{g=1}^{n_{\text{genes}}} y_{cg}$, $y_{\text{normalized}} = 25000$. Second, we divide the normalized data into intervals of unit 1 to round the data. Finally, to eliminate the impact of low-expression genes on the inference process, we filter out genes whose average gene expression is less than 1. After data processing, there are 730 genes from 593 cells left in control data set, 174 genes from 593 cells left in our 10 μm 5FU treatment data set and 187 genes from 564 cells left in 50 μm 5FU treatment data set.

Data availability

All the scRNA-seq data can be obtained from the public database. The scRNA-seq data of mouse embryonic stem cells (mESC) + DMSO and mESC + IdU data reported here is available at the Gene Expression Omnibus (GEO) database under accession number: GSE176044. The scRNA-seq data of RKO cell with different 5-fluorouracil treatment is available at the Gene Expression Omnibus (GEO) database under accession number: GSE149224.

Code availability

The code for reproducing the presented analysis results is available at GitHub repository (<https://github.com/cellfate/DeepTX>).

Additional information

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Author contributions

J.Z., Q.N., and B.J. conceived the project. J.Z., Z.H., and S.L. designed the algorithm. Z.H. and S.L. implemented the code, performed the simulations, analyzed the data, and interpreted the results. Z.W. and Z.Z. analyzed the data and interpreted the results. Q.N. and B.J. supported the data interpretation. Z.H., S.L., and J.Z. wrote the paper. All authors revised and approved the paper. J.Z. supervised the project.

Additional files

Supplemental File 

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Joint Public Review:

In this work, the authors present DeepTX, a computational tool for studying transcriptional bursting using single-cell RNA sequencing (scRNA-seq) data and deep learning. The method aims to infer transcriptional burst dynamics-including key model parameters and the associated steady-state distributions-directly from noisy single-cell data. The authors apply DeepTX to datasets from DNA damage experiments, revealing distinct regulatory patterns: IdU treatment in mouse stem cells increases burst size, promoting differentiation, while 5FU alters burst frequency in human cancer cells, driving apoptosis or survival depending on dose. These findings underscore the role of burst regulation in mediating cell fate responses to DNA damage.

The main strength of this study lies in its methodological contribution. DeepTX integrates a non-Markovian mechanistic model with deep learning to approximate steady-state mRNA distributions as mixtures of negative binomial distributions, enabling genome-scale parameter inference with reduced computational cost. The authors provide a clear discussion of the framework's assumptions, including reliance on steady-state data and the inherent unidentifiability of parameter sets, and they outline how the model could be extended to other regulatory processes.

The revised manuscript addresses many of the original concerns, particularly regarding sample size requirements, distributional assumptions, and the biological interpretation of inferred parameters. However, the framework remains limited by the constraints of snapshot data and cannot yet resolve dynamic heterogeneity or causality. The manuscript would also benefit from a broader contextualisation of DeepTX within the landscape of existing tools linking mechanistic modelling and single-cell transcriptomics. Finally, the interpretation of pathway enrichment analyses still warrants clarification.

Overall, this work represents a valuable contribution to the integration of mechanistic models with high-dimensional single-cell data. It will be of interest to researchers in systems biology, bioinformatics, and computational modelling.

<https://doi.org/10.7554/eLife.100623.2.sa1>

Author response:

The following is the authors' response to the original reviews

Joint Public Review:

In this work, the authors develop a new computational tool, DeepTX, for studying transcriptional bursting through the analysis of single-cell RNA sequencing (scRNA-seq) data using deep learning techniques. This tool aims to describe and predict the transcriptional bursting mechanism, including key model parameters and the steady-

state distribution associated with the predicted parameters. By leveraging scRNA-seq data, DeepTX provides high-resolution transcriptional information at the single-cell level, despite the presence of noise that can cause gene expression variation. The authors apply DeepTX to DNA damage experiments, revealing distinct cellular responses based on transcriptional burst kinetics. Specifically, IdU treatment in mouse stem cells increases burst size, promoting differentiation, while 5FU affects burst frequency in human cancer cells, leading to apoptosis or, depending on the dose, to survival and potential drug resistance. These findings underscore the fundamental role of transcriptional burst regulation in cellular responses to DNA damage, including cell differentiation, apoptosis, and survival. Although the insights provided by this tool are mostly well supported by the authors' methods, certain aspects would benefit from further clarification.

The strengths of this paper lie in its methodological advancements and potential broad applicability. By employing the DeepTXSolver neural network, the authors efficiently approximate stationary distributions of mRNA counts through a mixture of negative binomial distributions, establishing a simple yet accurate mapping between the kinetic parameters of the mechanistic model and the resulting steady-state distributions. This innovative use of neural networks allows for efficient inference of kinetic parameters with DeepTXInferer, reducing computational costs significantly for complex, multi-gene models. The approach advances parameter estimation for high-dimensional datasets, leveraging the power of deep learning to overcome the computational expense typically associated with stochastic mechanistic models. Beyond its current application to DNA damage responses, the tool can be adapted to explore transcriptional changes due to various biological factors, making it valuable to the systems biology, bioinformatics, and mechanistic modelling communities. Additionally, this work contributes to the integration of mechanistic modelling and -omics data, a vital area in achieving deeper insights into biological systems at the cellular and molecular levels.

We thank the reviewers for their positive opinion on our manuscript. As reflected in our detailed responses to the reviewers' comments, we will make significant changes to address their concerns comprehensively.

This work also presents some weaknesses, particularly concerning specific technical aspects. The tool was validated using synthetic data, and while it can predict parameters and steady-state distributions that explain gene expression behaviour across many genes, it requires substantial data for training. The authors account for measurement noise in the parameter inference process, which is commendable, yet they do not specify the exact number of samples required to achieve reliable predictions. Moreover, the tool has limitations arising from assumptions made in its design, such as assuming that gene expression counts for the same cell type follow a consistent distribution. This assumption may not hold in cases where RNA measurement timing introduces variability in expression profiles.

Thank reviewers for detailed and constructive feedback on our work. We will address the key concerns raised from the following points:

(1) Clarification on the required sample size: We tested the robustness of our inference method on simulated datasets by varying the number of single-cell samples. Our results indicated that the predictions of burst kinetics parameters become accurate when the number of cells reaches 500 (Supplementary Figure S3d, e). This sample size is smaller than the data typically obtained with current single-cell RNA sequencing (scRNA-seq) technologies, such as 10x Genomics and Smart-seq3 (Zheng GX et al., 2017; Hagemann-Jensen M et al., 2020). Therefore, we believed that our algorithm is well-suited for inferring burst kinetics from existing scRNA-seq datasets, where the sample size is sufficient for reliable predictions. We will clarify this point in the main text to make it easier for readers to use the tool.

(2) Assumption-related limitations: One of the fundamental assumptions in our study is that the expression counts of each gene are independently and identically distributed (i.i.d.) among cells, which is a commonly adopted assumption in many related works (Larsson AJM et al., 2019; Ochiai H et al., 2020; Luo S et al., 2023). However, we acknowledged the limitations of this assumption. The expression counts of the same gene in each cell may follow distinct distributions even from the same cell type, and dependencies between genes could exist in realistic biological processes. We recognized this and will deeply discuss these limitations from assumptions and prospect as an important direction for future research.

The authors present a deep learning pipeline to predict the steady-state distribution, model parameters, and statistical measures solely from scRNA-seq data. Results across three datasets appear robust, indicating that the tool successfully identifies genes associated with expression variability and generates consistent distributions based on its parameters. However, it remains unclear whether these results are sufficient to fully characterise the transcriptional bursting parameter space. The parameters identified by the tool pertain only to the steady-state distribution of the observed data, without ensuring that this distribution specifically originates from transcriptional bursting dynamics.

We appreciate reviewers' comments and the opportunity to clarify our study's contributions and limitations. Although we agree that assessing whether the results from these three realistic datasets can represent the characterize transcriptional burst parameter space is challenging, as it depends on data property and conditions in biology, we firmly believe that DeepTX has the capacity to characterize the full parameter space. This believes stems from the extensive parameters and samples we input during model training and inference across a sufficiently large parameter range (Method 1.3). Furthermore, the training of the model is both flexible and scalable, allowing for the expansion of the transcriptional burst parameter space as needed. We will clarify this in the text to enable readers to use DeepTX more flexibly.

On the other hand, we agree that parameter identification is based on the steady-state distribution of the observed data (static data), which loses information about the fine dynamic process of the burst kinetics. In principle, tracking the gene expression of living cells can provide the most complete information about real-time transcriptional dynamics across various timescales (Rodriguez J et al., 2019).

However, it is typically limited to only a small number of genes and cells, which could not investigate general principles of transcriptional burst kinetics on a genome-wide scale. Therefore, leveraging the both steady-state distribution of scRNA-seq data and mathematical dynamic modelling to infer genome-wide transcriptional bursting dynamics represents a critical and emerging frontier in this field. For example, the statistical inference framework based on the Markovian telegraph model, as demonstrated in (Larsson AJM et al., 2019), offers a valuable paradigm for understanding underlying transcriptional bursting mechanisms. Building on this, our study considered a more generalized non-Mordovian model that better captures transcriptional kinetics by employing deep learning method under conditions such as DNA damage. This provided a powerful framework for comparative analyses of how DNA damage induces alterations in transcriptional bursting kinetics across the genome. We will highlight the limitations of current inference using steady-state distributions in the text and look ahead to future research directions for inference using time series data across the genome.

A primary concern with the TXmodel is its reliance on four independent parameters to describe gene state-switching dynamics. Although this general model can capture specific cases, such as the refractory and telegraph models, accurately estimating the

parameters of the refractory model using only steady-state distributions and typical cell counts proves challenging in the absence of time-dependent data.

We thank reviewers for highlighting this critical concern regarding the TXmodel's reliance on four independent parameters to describe gene state-switching dynamics. We acknowledge that estimating the parameters of the TXmodel using only steady-state distributions and typical single-cell RNA sequencing (scRNA-seq) data poses significant challenges, particularly in the absence of timeresolved measurements.

As described in the response of last point, while time-resolved data can provide richer information than static scRNA-seq data, it is currently limited to a small number of genes and cells, whereas static scRNA-seq data typically capture genome-wide expression. Our framework leverages deep learning methods to link mechanistic models with static scRNA-seq data, enabling the inference of genome-wide dynamic behaviors of genes. This provides a potential pathway for comparative analyses of transcriptional bursting kinetics across the entire genome.

Nonetheless, the refractory model and telegraphic model are important models for studying transcription bursts. We will discuss and compare them in terms of the accuracy of inferred parameters.

Certainly, we agree that inferring the molecular mechanisms underlying transcriptional burst kinetics using time-resolved data remains a critical future direction. We will include a brief discussion on the role and importance of time-resolved data in addressing these challenges in the discussion section of the revised manuscript.

The claim that the GO analysis pertains specifically to DNA damage response signal transduction and cell cycle G2/M phase transition is not fully accurate. In reality, the GO analysis yielded stronger p-values for pathways related to the mitotic cell cycle checkpoint signalling. As presented, the GO analysis serves more as a preliminary starting point for further bioinformatics investigation that could substantiate these conclusions. Additionally, while GSEA analysis was performed following the GO analysis, the involvement of the cardiac muscle cell differentiation pathway remains unclear, as it was not among the GO terms identified in the initial GO analysis.

We thank the reviewer for this valuable feedback and for pointing out the need for clarification regarding the GO and GSEA analyses. We agree that the connection between the cardiac muscle cell differentiation pathway identified in the GSEA analysis and the GO terms from the initial analysis requires further clarification. This discrepancy arises because GSEA examines broader sets of pathways and may capture biological processes not highlighted by GO analysis due to differences in the statistical methods and pathway definitions used. We will revise the manuscript to address this point, explicitly discussing the distinct yet complementary nature of GO and GSEA analyses and providing a clearer interpretation of the results.

As the advancement is primarily methodological, it lacks a comprehensive comparison with traditional methods that serve similar functions. Consequently, the overall evaluation of the method, including aspects such as inference accuracy, computational efficiency, and memory cost, remains unclear. The paper would benefit from being contextualised alongside other computational tools aimed at integrating mechanistic modelling with single-cell RNA sequencing data. Additional context regarding the advantages of deep learning methods, the challenges of analysing large, high-dimensional datasets, and the complexities of parameter estimation for intricate models would strengthen the work.

We greatly appreciate your insightful feedback, which highlights important considerations for evaluating and contextualizing our methodological advancements. Below, we emphasize our advantages from both the modeling perspective and the inference perspective compared with previous model. As our work is rooted in a model-based approach to describe the transcriptional bursting process underlying gene expression, the classic telegraph model (Markovian) and non-Markovian models which are commonly employed are suitable for this purpose:

Classic telegraph model: The classic telegraph model allows for the derivation of approximate analytical solutions through numerical integration, enabling efficient parameter point estimation via maximum likelihood methods, e.g., as explored in (Larsson AJM et al., 2019). Although exact analytical solutions for the telegraph model are not available, certain moments of its distribution can be explicitly derived. This allows for an alternative approach to parameter inference using moment-based estimation methods, e.g., as explored in (Ochiai H et al., 2020). However, it is important to note that higher-order sample moments can be unstable, potentially leading to significant estimation bias.

Non-Markovian Models: For non-Markovian models, analytical or approximate analytical solutions remain elusive. Previous work has employed pseudo-likelihood approaches, leveraging statistical properties of the model's solutions to estimate parameters ,e.g., as explored in (Luo S et al., 2023).

However, the method may suffer from low inference efficiency.

In our current work, we leverage deep learning to estimate parameters of TXmodel, which is nonMarkovian model. First, we represent the model's solution as a mixture of negative binomial distributions, which is obtained by the deep learning method. Second, through integration with the deep learning architecture, the model parameters can be optimized using automatic differentiation, significantly improving inference efficiency. Furthermore, by employing a Bayesian framework, our method provides posterior distributions for the estimated dynamic parameters, offering a comprehensive characterization of uncertainty. Compared to traditional methods such as moment-based estimation or pseudo-likelihood approaches, we believe our approach not only achieves higher inference efficiency but also delivers posterior distributions for kinetics parameters, enhancing the interpretability and robustness of the results. We will present and emphasize the computational efficiency and memory cost of our methods the revised version.

Recommendations for the authors:

There are various noise sources in biological progress. How transcriptional bursting fits within those as well as the reasons to focus only on this source needs to be clearly discussed in the introduction of the manuscript. Related to this last point, transcriptional bursting might not be the only mechanism to take advantage of the stochastic nature of biomolecular processes to make decisions. Once again, what are the implications of assuming this as the underlying mechanism?

Thank the reviewer for this valuable comment. We fully agree that biological systems are subject to multiple stochastic sources, which arise from both intrinsic and extrinsic noise (Eling N et al., 2019). Intrinsic noise is primarily driven by the stochastic biochemical effects that directly influence mRNA and protein expression in a gene-specific manner, such as DNA, epigenetic, transcription, and translation levels. Extrinsic noise arises from fluctuations in cell-specific manners, such as changes in cell size, cell cycle, or cell signaling. Given that DNA damage most directly perturbs transcription and translation processes, focusing on intrinsic noise sources is appropriate for mechanistically modeling gene-specific expression

variability, particularly since this variability can be captured at the genome-wide scale by scRNA-seq data.

Among various intrinsic noise sources, transcriptional bursting offers a mechanistically wellcharacterized and quantifiable representation of gene expression variability (Tunnacliffe E & Chubb JR, 2020). It reflects the dynamic switching between active and inactive gene states and has been observed consistently across prokaryotic and eukaryotic cells (Eling N et al., 2019). Moreover, transcriptional bursting kinetics, defined by burst size and frequency, can be inferred from scRNA-seq data at the singlegene level using steady-state assumptions, making it an analytically tractable and biologically meaningful feature for large-scale inference (Rodriguez J & Larson DR, 2020).

We acknowledge that transcriptional bursting is not the only mechanism through which cells can utilize stochasticity for fate decisions. Other processes, such as translational noise and chromatin accessibility, may also contribute. However, given the data modality (static scRNA-seq) and the established theoretical framework for bursting, we assume transcriptional bursting as a representative and interpretable proxy of stochastic regulation. This assumption enables us to extract meaningful insights while remaining open to future model extensions, incorporating additional regulatory layers as more data types become available.

In this version of the manuscript, we have revised the introduction section to better clarify the rationale of this assumption and to more explicitly emphasize the important role of transcriptional bursting within stochastic noise.

More careful discussion of how the proposed method differentiates from previous work that employs scRNA-seq to elucidate the diverse sources of noise (pp.3).

Thank the reviewer for this suggestion. Our proposed method differs significantly from previous work that utilizes scRNA-seq data to study diverse noise sources from several aspects (Ochiai H et al., 2020; Eling N et al., 2019; Morgan MD & Marioni JC, 2018). Specifically, DeepTX infers genomewide burst kinetics by directly matching the full steady-state distribution of a mechanistic stochastic model to the observed scRNA-seq data, rather than relying solely on low-order statistics such as mean and variance. Moreover, by adopting a non-Markovian process that allows multi-step promoter switching, DeepTX extends beyond the classic telegraph model to better capture the complex molecular events underlying transcriptional activation and repression. Crucially, we used a deep-learning-based solver to obtain these intractable steady-state distributions rapidly and accurately. This combination of richer data usage, more realistic mechanistic assumptions, and scalable neural-network-accelerated computation lays the groundwork for incorporating additional noise sources into a unified inference framework in future work.

In this version of the manuscript, we have revised the discussion section to highlight the difference with previous works.

The paper could benefit from being contextualised alongside other computational tools that aim to integrate mechanistic modelling with single-cell RNA sequencing data. This is an active area of research, and works such as Sukys and Grima (bioRxiv, 2024), Garrido-Rodriguez et al. (PLOS Computational Biology, 2021), Maizels (2024), and others could provide valuable context.

Thank the reviewer for suggesting these relevant works. Garrido-Rodriguez et al. (PLOS Comput. Biol., 2021) integrated single-cell and bulk transcriptomic data into mechanistic pathway models to infer signaling dynamics, an approach complementary to our mapping of burst kinetic parameters onto pathway enrichment for linking transcriptional bursting to functional outcomes. Sukys and Grima et al. (bioRxiv, 2024; Now in Nucleic Acids Res., 2025)

demonstrated that cell-cycle stage and cellular age significantly modulate burst frequency and size, highlighting the potential to enhance DeepTX by incorporating cell-cycle-dependent variability into genome-wide burst inference. Maizels et al. (Philos. Trans. R. Soc. Lond. B. Biol. Sci., 2024) reviewed methods for capturing single-cell temporal dynamics across multi-omic modalities, underscoring how higher time-resolved data could refine and validate steady-state burst inference frameworks to better resolve causal gene-expression mechanisms.

We have cited these studies on the contextual relevance to DeepTX in the discussion sections.

As the advancement is primarily methodological, it lacks a comprehensive comparison with traditional methods that serve similar functions. Consequently, the overall evaluation of the method, including aspects such as inference accuracy, computational efficiency, and memory cost, remains unclear. We suggest incorporating these experiments to provide readers with a more complete understanding of the proposed method's performance.

Thank the reviewer for constructive suggestion regarding a comprehensive comparison with other previous methods. To address this problem, in this version, we compared DeepTX with our previous work, txABC, that utilized approximate Bayesian computation to infer parameters from the generalized telegraph model (Luo S et al., 2023). As a result, DeepTX achieved improvements in inference accuracy and computational efficiency (Supplementary Figure S4.). For memory cost during single-gene inference, DeepTX requires an average memory usage of approximately 70 MB, whose memory consumption accounts for only a small fraction of the total available memory on standard computing devices (typically exceeding 10 GB), while exhibiting superior inference efficiency compared to txABC. We have mentioned in the third result section.

Discuss the validity of the assumption of the static snapshot provided by the scRNA-seq data as in steadystate (i.e., stationary distribution), and the implications of this assumption being untrue (for the proposed method).

We thank the reviewer for the comment regarding the stationary assumption. We assume that each scRNA-seq snapshot approximates the steady-state (stationary) distribution of transcript counts because (i) typical single-cell experiments sample large, asynchronously dividing populations that collectively traverse many transcriptional burst cycles, and (ii) in the absence of a synchronized perturbation, mRNA production and degradation reach a dynamic balance on timescales much shorter than overall cell-type changes. Under these conditions, the empirical count distribution closely mirrors the model's stationary solution, justifying steady-state inference of burst size and frequency from a single time point. This assumption is commonly adopted in probabilistic models of transcriptional bursting (Larsson AJM et al., 2019; Raj A & van Oudenaarden A, 2008).

However, this steady-state assumption has some limitations. First, in some scenarios, the cell system may exhibit highly transient transcriptional programs that do not satisfy stationarity, leading to biased or misleading parameter estimates. For example, immediately following a synchronized developmental stimulus—such as serum shock-induced activation of immediate-early genes. Second, because DeepTX infers the mean burst frequency and size across the population, it cannot recover the underlying time-resolved dynamics or distinguish heterogeneous kinetic subpopulations.

We have added a statement in the discussion to acknowledge these limitations and suggest future extensions—such as incorporating time-series measurements or latent pseudo time covariates—to address non-stationarity and recover temporal burst dynamics.

On page 3, "traditional telegraph model" is mentioned without any context. This model, and particularly the implications for the current work, might not be obvious to the reader. Take one or two sentences to give the reader context.

Thank the reviewer for this helpful comment. We acknowledge that the mention of the "traditional telegraph model" on page 3 may not be immediately clear to all readers. The traditional telegraph model is a mathematical framework commonly used to describe gene expression burst dynamics, in which genes stochastically switch between active (ON) and inactive (OFF) states, with exponentially distributed waiting times for state transitions. To provide the necessary context, we added a brief introduction to the traditional telegraph model and its relevance to our work in the revised manuscript.

A primary concern with the model used in Figure 2a (TXmodel) is its reliance on four independent parameters to describe gene state switching dynamics. While this general model can encompass specific cases such as the refractory model (Science 332, 472 (2011)) and the telegraph model, accurately estimating the parameters of the refractory model using only steady-state distributions and typical cell numbers (10^3 - 10^4) is challenging without time-dependent data. To address this, we suggest that the authors provide parameter inference results for each individual parameter, rather than only for burst size and burst frequency, based on synthetic data. This would help clarify the model's effectiveness and improve understanding of its estimation precision.

Thank the reviewer for highlighting this important concern. We agree that the lack of timeresolved measurements may affect the accuracy of inferences about dynamic parameters, especially the unidentifiability of parameters inferred from steady-state distributions, i.e., multiple parameters leading to the same steady-state distribution. The unidentifiability of individual parameters is a common and critical problem in systems biology studies. To address this issue, for example, Trzaskoma et al. developed StochasticGene, a computationally efficient software suite that uses Bayesian inference to analyze arbitrary gene regulatory models and quantify parameter uncertainty across diverse data types (Trzaskoma P et al., 2024). Alexander et al. adopt a Bayesian approach to parameter estimation by incorporating prior knowledge through a prior distribution and classify a parameter as practically nonidentifiable if it cannot be uniquely determined beyond the confidence already provided by the prior (Browning AP et al., 2020). Hence, in DeepTX, we employed a Bayesian approach based on loss potential to infer the posterior distributions of the parameters (Figure 3E).

Although DeepTX also encounters the issue of unidentifiability for individual parameters (Supplementary Figure S11), the multimodal nature of the posterior distribution suggests that multiple distinct parameter sets can produce similarly good fits to the observed data, highlighting the inherent non-identifiability of the model. Nevertheless, in the multimodal posterior distribution, at least one of the posterior peaks aligns closely with the ground truth, thereby demonstrating the validity of the inferred result. Moreover, inference results on synthetic data confirm that the BS and BF can be accurately estimated (Supplementary Figure S3b and S3c). We also performed robustness analyses on synthetic datasets. As shown in Supplementary Figure S3d and S3e, our model reliably recovers the ground-truth burst kinetics of models when the number of cells reaches ~1000, which is within the range of typical single-cell RNA-seq experiments.

We have explicitly pointed out the potential issue of unidentifiability due to the lack of temporal resolution information in the discussion section.

Noteworthy, transcriptional is always a multi-step process (depending on the granularity with which the process is described). What do the authors mean by saying that "DNA

damage turns transcription into a multi-step process rather than a single-step process”?

Thank the reviewer for pointing out the lack of precision in our original statement. We agree that the phrasing could be misleading. Transcription is inherently a multi-step process, but most mechanistic studies simplify it to a single-step “telegraph” model for tractability. In the context of DNA damage, however, damage-induced pausing and repair-mediated delays introduce additional intermediary states in the transcription cycle that cannot be approximated by a single step. To capture these damage-specific interruptions, DeepTX explicitly consider a multi-step promoter switching framework rather than combining all transitions into one. What we originally wanted to express was the necessity of multi-step process modeling. We have replaced the original sentence in introduction with: “However, the presence of DNA damage necessitates modeling the transcriptional process as a multistep process, rather than a single-step process, to capture the additional complexity introduced by the damage”.

*It is unclear why the authors have chosen a different definition in Equation (2) rather than the commonly used burst frequency, $1/(k_{\text{deg}} * \tau_{\text{off}})$, as reported in the literature. Unlike the traditional definition, which is unit-free, the definition in Eq. (2) includes units, raising questions about its interpretability and consistency with established conventions. Clarifying this choice would improve the understanding and consistency of the methodology.*

Thank the reviewer for raising this important point. We acknowledge that there are multiple definitions of burst frequency (BF) in the literature. Here, we provide a detailed explanation, clarifying the differences between these definitions, including the one used $1/(\langle \tau_{\text{off}} \rangle + \langle \tau_{\text{on}} \rangle)$ and the traditional definition $1/(k_{\text{deg}} * \tau_{\text{off}})$.

First, the definition of burst frequency we adopt has been widely used in recent literatures, such as Benjamin Zoller et al. (Zoller B et al., 2018), Caroline Hoppe et al. (Hoppe C et al., 2020) and Daniel Ramsköld (Ramsköld D et al., 2024). And its quantity represents the average time it takes for the promoter to complete one full stochastic cycle between its active and inactive states $1/(\langle \tau_{\text{off}} \rangle + \langle \tau_{\text{on}} \rangle)$. Secondly, the traditional definition $1/(k_{\text{deg}} * \tau_{\text{off}})$ can be regarded as a simplified version of our definition, under the assumptions that τ_{on} is negligible and $k_{\text{deg}} = 1$ (i.e., rate parameters are normalized to be unit-free). Although it is reasonable to neglecting activate time τ_{on} , as it is typically much shorter than inactive time under some conditions, we chose a more complete way to define the burst frequency so that it is applicable to more general situations. In addition, by defining the burst frequency as $1/(\langle \tau_{\text{off}} \rangle + \langle \tau_{\text{on}} \rangle)$, the mean transcription level can be analytically represented as the product of burst size and burst frequency. This explanation has been clarified in the methods 1.2 section.

The authors mention the need to model "more realistic gene expression processes". How is this exactly being incorporated into the model?

Thank the reviewer for raising this important question. To incorporate "more realistic gene expression processes" into our model, we considered two critical aspects into DeepTX that are often oversimplified in traditional approaches:

(1) Integration of gene expression and sequencing processes: Observations from scRNA-seq data are influenced by both the intrinsic gene expression processes and the subsequent sequencing procedure. Traditional models often focus solely on gene expression, neglecting the stochastic effects introduced by the sequencing process. Our model explicitly incorporates both the gene expression and sequencing processes, providing a more comprehensive and realistic representation of the observed data.

(2) Modeling gene expression as a multi-step process: Gene expression is inherently a multi-step process. However, traditional telegraph models typically simplify gene state switching as a single-step process for tractable analysis, often assuming Markovian dynamics where transition waiting times follow exponential distributions. In contrast, our model accounts for the multi-step nature of gene state transitions by allowing the waiting times to follow non-exponential (non-Markovian) distributions. This model is more suitable for gene expression dynamics that cannot be simplified to a single-step process, such as DNA damage, which may introduce an intermediate state to represent pausing and repair in the transcription process.

By addressing these factors, our model better reflects the complexity and stochastic nature of gene expression processes, aligning more closely with the data generated from biological systems. We have added detailed explanations after this sentence for clarification in the first result section.

Better explanation of the previously developed TXmodel, and the assumption of a non-Markovian system. In particular, it isn't clear how using arbitrary distributions for the waiting times implies a non-Markovian process (as the previous state(s) of the system is not used to inform the transition probability, at least as explained in pp. 4). Without a clear discussion of the so-called arbitrary waiting time distribution, it isn't clear how these represent a mechanistic model. In general, a more careful discussion of the "mechanistic" model is needed.

Thank the reviewer for this thoughtful comment. In this revised version, we provided a more detailed explanation of the relationship between the TXmodel and the non-Markovian system in the revised manuscript. Specifically, we will clarify the following points:

(1) Why non-Markovian system: In a Markovian system, the waiting times for events are exponentially distributed, meaning that the state transitions depend solely on the current state and are memoryless (Van Kampen NG, 1992). However, when the waiting times follow non-exponential distributions, such as Gamma or Weibull distributions, the state transitions are no longer independent of the system's previous states. This introduces memory into the system, making it non-Markovian.

(2) Why mechanistic model: First, it is important to clarify that regardless of whether the waiting time is arbitrary or exponential (corresponding to non-Markovian and Markovian systems), our TXmodel is a mechanistic model because it models the dynamic process of transcription bursts with interpretable kinetic parameters. Second, although we introduced arbitrarily distributed waiting times, reasonable selection of waiting time distributions can still make the distribution parameters mechanistically interpretable. For example, in the context of modeling ON and OFF state switching times using a Gamma distribution, the two parameters have clear interpretations: the shape parameter represents the number of sequential exponential (memoryless) steps required for the transition to occur, capturing the complexity or multi-step nature of the switching process, while the scale parameter denotes the average duration of each of these steps. We have added the explanation in methods 1.2 section.

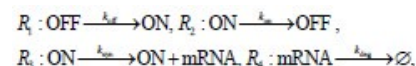
Include a brief discussion about the metric used to compare distributions (and introduce KL abbreviation).

Thank the reviewer for this suggestion. In the second result and methods 1.3 section of revised manuscript, we have included a brief discussion to introduce and clarify the metric used to compare distributions. Specifically, we have given more explanation for the Kullback-Leibler (KL) divergence, which is a widely used metric for quantifying the difference between two probability distributions. We also ensured that the abbreviation "KL" is properly introduced when it first appears in the text, along with a concise description of its mathematical definition and interpretation within the context of our analysis.

| *What does the "CTM" model stand for (in supplementary information)? And "TX" model?*

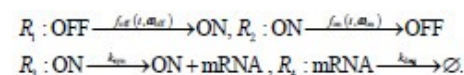
Thank the reviewer for highlighting this point. We revised the supplementary information to explicitly define the "CTM" and "TX" models and clarify their distinctions.

CTM model: The "CTM" model refers to the classic telegraph model, a widely used model for capturing Markovian gene expression burst kinetics. The CTM describes stochastic gene expression as a sequence of four biochemical reactions involving two gene states (ON and OFF), mRNA transcription and degradation:



k_{off} as the rate at which the gene switches from OFF to ON, k_{on} as the rate at which the gene switches from ON to OFF, k_{syn} as the rate of mRNA synthesis and k_{deg} as the rate of mRNA degradation. In this model, gene switching between active and inactive states is governed by a memoryless Markovian process, where the waiting times for transitions follow exponential distributions (Van Kampen NG, 1992).

TX model: In contrast, the "TX" model is a more generalized telegraph model for transcriptional processes.



Different from the CTM, the waiting times for state transitions between ON and OFF in the TX model follow arbitrary waiting time distributions. This implies that the future state of the system depends not only on the current state but may also be influenced by its historical trajectories. Consequently, the TX model exhibits non-Markovian behavior. We have added more detailed description on these two models in section 1.1 of supplementary text.

| *Leaky transcription (in the OFF promoter state) is not considered. What would be the implications of its presence in the data?*

Thank the reviewer for pointing out the potential role of leaky transcription in our analysis. We acknowledge that leaky transcription, occurring in the promoter OFF state, was not explicitly considered in our current model. Our decision to exclude it assumed that the leaky transcription rate is relatively small and its impact on the observed data is negligible. This assumption is consistent with previous studies that similarly disregard leaky transcription in gene expression modeling due to its minimal contribution to the overall dynamics (Larsson AJM et al., 2019).

However, we recognize that the leaky transcription should be considered, particularly in systems where the leaky rate is significant relative to the active transcription rate. In such

cases, it may introduce additional variability to the observed expression levels or obscure the distinction between ON and OFF states. We have added relevant statements in the discussion section.

In the main text, the waiting time for state transitions is described by two parameters, while in the methods/supplementary information only one parameter is considered per distribution (without a clear discussion of the so-called "dwell time distributions").

Thank the reviewer for this comment. We recognize the need to clarify the discrepancy between the descriptions of waiting times in the main text and supplementary materials.

Dwell time distribution refers to the probability distribution of the time in which a gene remains in a particular transcriptional state (ON or OFF) before transitioning to the other state. While in Markovian models the dwell time follows an exponential distribution, more complex or non-Markovian regulatory mechanisms may give rise to Gamma, Weibull, or other non-exponential dwell time distributions.

In our model, we denote the dwell time distributions in the OFF and ON states by $f_{\text{off}}(t, \omega_{\text{off}})$ and

$f_{\text{on}}(t, \omega_{\text{on}})$, respectively, where w represents a vector of parameters characterizing the

distribution, the dimensionality of which depends on the specific form of the distribution. For example, when an exponential distribution is assumed, w consists of a single rate parameter; in contrast, for distributions such as the Gamma or Weibull, w includes two parameters. In

the main text, both $f_{\text{off}}(t, \cdot)$ and $f_{\text{on}}(t, \cdot)$ are modeled using Gamma distributions, whereas in the

Supplementary Materials, we assume exponential distributions for both, resulting in a single-parameter representation. We have added relevant statements in the methods 1.2 section.

Related, but more general, across the manuscript there are problems with the consistency in terminology. This is especially problematic with the figures. It makes it incredibly hard to follow the work. Better integration of the information, and consistency with the terminology, would improve the understanding for the reader.

Thank the reviewer for the valuable feedback. To enhance clarity and readability, we have carefully revised the manuscript to ensure consistent terminology throughout the text and figures e.g., unifying terms such as "untreatment" and "control" under the consistent label "control"—across both the text and figures.

One of the four main assumptions behind the model is that "the solution of the model can be explained by a mixed negative binomial distribution". The logic and implications of this assumption need to be discussed in the paper. (Methods, pp.13.) All four assumptions need to be carefully argued in the paper.

We appreciate the reviewer's comment regarding the assumptions underlying our model. Here, we would like to clarify the rationale and implications of each assumption.

Assumption 1 (The gene expression of cells was in a stationary distribution during sequencing.) has been extensively used in previous studies for the inference and modeling of scRNA-seq data, demonstrating effectiveness in capturing mRNA expression distributions and inferring underlying dynamic parameters (Larsson AJM et al., 2019; Luo S et al., 2023; Ramsköld D et al., 2024; Gupta A et al., 2022).

For Assumption 2 (Gene expression counts of the same cell type follow the same distribution.) is as follows: cell types are typically defined based on gene expression profiles or functional characteristics. Cells with similar functions often exhibit consistent transcriptional programs, leading to approximately identical gene expression distributions. This assumption has been widely adopted in previous research (Larsson AJM et al., 2019; Gupta A et al., 2022).

Regarding Assumption 3 (The solution of the model can be approximated by a mixed negative binomial distribution.), in the most general formulation, a chemical master equation (CME) model of biological systems converges to a stationary distribution $P(n; \theta)$ over $n \in \mathbb{N}$. And

$P(n; \theta)$ afford a real Poisson representation (Gardiner CW & Chaturvedi S, 1977): $P(n; \theta) = \int_0^\infty P_{\text{Poisson}}(n; \lambda) dF_\lambda$

where F is a mixing cumulative distribution function (CDF). If such a Poisson representation exists, we can always write down a finite approximation over K Poisson kernels:

$P(n; \theta) = \sum_{k=1}^K w_k P_{\text{Poisson}}(n; \lambda_k)$, where w_k are weights on a K -dimensional simplex. Further, as k

$\rightarrow \infty, Q \rightarrow P$. More problematically, convergence in the number of kernels in K is typically slow. Negative binomial kernels $P_{\text{Poisson}}(n; m_k, l_k)$, which are continuous Poisson mixtures with a gamma mixing density can accelerate convergence in K (Gorin G et al., 2024). Hence, the solution of the TX model can be approximated by a mixed negative binomial distribution.

For Assumption 4 (The state space sampled from a sufficiently long single simulation is statistically equivalent to that obtained from multiple simulations at steady state in gene expression models.), when a sample trajectory of the model is simulated for a sufficiently long period, it is assumed to have traversed the entire stationary state space (Kuntz J et al., 2021). Therefore, by performing truncated statistical analysis on the trajectory, the corresponding stationary distribution of the model can be obtained. We have added the explanation in methods 1.1 section.

The authors propose that the waiting times between promoter states follow a non-exponential distribution, but the choice of gamma distribution and the implications for the method and the biological conclusions need to be discussed.

We thank the reviewer for this comment. To account for the impact of DNA damage on the transcription process, our model assumes that both the "ON" and "OFF" states of the promoter consist of multiple underlying sub-states. When a promoter switches from the "ON" state to the "OFF" state, the transition is governed by multiple distinct waiting time distributions that follow exponential distributions. Similarly, when a promoter switches from the "OFF" state to the "ON" state, there may be multiple transitions from different "OFF" sub-states. Consequently, the waiting times for the transitions from the "OFF" state to the "ON" state, and vice versa, must account for multiple exponential waiting time distributions associated with each "ON" state transition. We can map a multiple exponential-waiting-times reaction process to a single-step reaction process with a non-exponential waiting time distribution. Therefore, we use a Gamma distribution for dwell time of promoter switching, which can be expressed as the convolution of multiple exponential distributions (corresponding to a sum of multiple exponential variables). Additionally, other non-exponential distributions, such as those discussed in our previous studies (Zhang J & Zhou T, 2019), may also be considered, and we recognize that alternative choices could be made depending on the specific characteristics of the system. We have added the explanation in methods 1.2 section.

*BF - burst frequency; BS - burst size. These terms represent the main data output, but they are only mathematically defined in the methods, and never the intuition of the specific expression explained (e.g., why not using $t_{\text{ON}}/(t_{\text{ON}}+t_{\text{OFF}})$ as BF instead of $1/(t_{\text{ON}}+t_{\text{OFF}})$, and why not $k_{\text{SYN}}*t_{\text{ON}}$ as BS instead of $k_{\text{SYN}}*t_{\text{ON}}$).*

We appreciate the reviewer's comment and agree that clarifying the biological intuition behind the mathematical definitions of burst frequency (BF) and burst size (BS) is important. Below, we provide a more detailed explanation of these definitions.

BF: The definition of burst frequency $1/(\langle\tau_{\text{off}}\rangle+\langle\tau_{\text{on}}\rangle)$ we adopt has been widely used in previous literature, such as Benjamin Zoller et al (Zoller B et al., 2018), Caroline Hoppe et al (Hoppe C et al., 2020) and Daniel Ramsköld (Ramsköld D et al., 2024). And its quantity represents the average time it takes for the promoter to complete one full stochastic cycle between its active and inactive states.

BS: The definition of burst size $BS = k_{\text{syn}}\langle\tau_{\text{on}}\rangle$ we adopt is consistent with the definition proposed by the reviewer. Burst size refers to the average number of mRNA transcripts produced during a single transcriptional activation event of a gene. It reflects the quantity of gene product synthesized per activation and is influenced by the rate of transcription and the duration of the active state of the gene. Our definition aligns with this biological interpretation and is mathematically formulated as $BS = k_{\text{syn}}\langle\tau_{\text{on}}\rangle$, where k_{syn} is the

transcription rate and $\langle\tau_{\text{on}}\rangle$ is the average duration of the active state.

In addition, the mean transcription level can be analytically represented as the product of burst size and burst frequency. This analytical result has been included in the methods 1.2 section of revised manuscript.

One can assume from the methods that omegaON and omegaOFF are the vector of (2) parameters describing the distribution, but the reader would benefit from some clarity here. The authors claim that they proved that the distribution moments can be obtained through an iterative process. How much does this rely on the assumption of an underlying binomial distribution?

Thank the reviewer for this helpful suggestion. To clarify, the vectors omegaON and omegaOFF represent the parameters characterizing the waiting time distributions of the promoter's active and inactive states, respectively. The exact form and interpretation of these vectors depend on the specific distributional choice for the waiting times. For instance, when the waiting time distribution follows a Gamma distribution with shape parameter $\alpha > 0$ and scale parameter $\beta > 0$, denoted as $f(x; \alpha, \beta) = x^{\alpha-1} e^{-x/\beta} / \Gamma(\alpha) \beta^\alpha, x \geq 0$, then $w_{\text{on}} = (\alpha, \beta)$. Conversely, when the waiting time distribution follows a Weibull distribution, denoted as

$f(x; \lambda, k) = k / \lambda (x / \lambda)^{k-1} e^{-(x/\lambda)^k}, x \geq 0$, with shape parameter $k > 0$ and scale parameter $\lambda > 0$, then $w_{\text{on}} = (l, k)$. We have clarified it in the Methods 1.2 section of the revised manuscript.

For the question about the binomial distribution, in our work, we use the binomial moment method to compute distributional statistics of chemical master equation (Zhang J et al., 2016).

Binomial moments of the mRNA stationary distribution $P(m)$ are defined as $b_n = \sum_{m=0}^n \binom{n}{m} P(m)$

where the symbol $\binom{n}{m}$ represents the combinatorial number. This technique refers to a

mathematical tool for moment calculation and is not based on the assumption that the underlying distribution is binomial distribution (Luo S et al., 2023). Hence, our approach is general and does not require the distribution itself to follow a binomial form.

More details about the parameter sampling are required. For instance, why are the specific ranges chosen and their implications? And is the space explored in logarithmic scale?

Thank the reviewer for the insightful comment regarding parameter sampling. In our study, we considered five parameters: $\theta = (k_{\text{off}}, r_{\text{off}}, k_{\text{on}}, r_{\text{on}}, k_{\text{syn}})$. The parameters k_{off} and k_{on} represent

the number of intermediate reaction steps involved in transcriptional state transitions. These values were sampled uniformly from the range 1 to 15, which aligns with biological evidence indicating that most genes undergo either direct (single-step) transitions or a small number of intermediate steps, typically fewer than ten (Tunnacliffe E & Chubb JR, 2020). This range is sufficient to capture both widely used singlestep models and more detailed multi-step mechanisms without introducing biologically implausible complexity.

Among these parameters, r_{off} and r_{on} denote the rate constants governing stochastic transitions between the OFF and ON transcriptional states, respectively. The mean duration of the OFF state, which corresponds to the time between transcriptional bursts, is given by

$\langle \tau_{\text{off}} \rangle = k_{\text{off}} / r_{\text{off}}$, and falls within the range $\langle \tau_{\text{off}} \rangle \in (0.1, 150)$. Experimental measurements report a

median value of $\langle \tau_{\text{off}} \rangle$ approximately 3.7 (Gupta A et al., 2022), which is well contained within

this range. Similarly, the mean duration of the ON state, referred to as the burst duration, is

defined by $\langle \tau_{\text{on}} \rangle = k_{\text{on}} / r_{\text{on}}$, and spans the interval $\langle \tau_{\text{on}} \rangle \in (0.1, 1500)$. The experimentally

observed median value of 0.12 (Gupta A et al., 2022) confirms that the parameter range adequately captures biologically realistic dynamics.

The parameter k_{syn} represents the normalized synthesis rate after accounting for molecular degradation. Its range was chosen based on empirical observations of transcriptional burst sizes, which typically vary from single molecules to several dozen (Gupta A et al., 2022).

Considering the relationship $BS = k_{\text{syn}} * \langle \tau_{\text{on}} \rangle$, the selected range of k_{syn} ensures that the

experimentally observed burst sizes are well represented within the defined parameter space. We have added the explanation in methods 1.2 section and supplementary text 4.

We fully recognize the advantages of logarithmic sampling, particularly when parameters span several orders of magnitude. Logarithmic scaling ensures balanced exploration across wide ranges and prevents sampling bias towards larger values. However, in our work, we applied Sobol sampling directly within the original (linear) parameter space. Although we did not explicitly transform parameters into logarithmic scale, Sobol sequences provide low-discrepancy, quasi-random coverage, which promotes uniform sampling across bounded domains (Sobol IM, 1967). Further, if necessary, we can increase the parameter range adaptively, and perform simulation algorithm to obtain sample and train a new model to solve a larger parameter range.

On page 15, the rationale for selecting the parameter space is unclear. This is crucial, as fully connected neural networks typically exhibit poor extrapolation beyond their training parameter space. If the parameter space of an experimental dataset significantly differs

from the training range, the inference results may become unreliable. We suggest further clarification on how the alignment between the parameter spaces of the experimental data and the training dataset can be ensured to maintain inference accuracy.

We appreciate the reviewer's insightful comment regarding the extrapolation limitations of fully connected neural networks. To address this concern, we have implemented a truncation strategy during inference, which constrains the inferred parameters to remain within the bounds of the training parameter space. This ensures that the neural network operates within a regime where its predictive accuracy has been validated, thereby enhancing the robustness of our results. Additionally, we have carefully selected the training parameter space to be reasonable, based on the characteristics of the experimental data. These ranges have been validated through domain knowledge and data analysis, ensuring that even when the experimental data approaches the boundaries of the training range, the inference results remain reliable and accurate.

On page 16, it is unclear why the authors chose to incorporate the Fano factor instead of using the coefficient of variation or variance. Clarifying the reasoning behind the selection of the Fano factor over these other statistical measures would provide better insight into its relevance for their analysis.

We thank the reviewer for raising this point. Although the loss term is described using the Fano factor, its formulation actually involves both the variance and the mean. Specifically,

the loss we use is: $(\log(\hat{s}_{\text{train}}/\hat{s}_{\text{mean}}))^2 = (\log((\hat{s}_{\text{var}}/\hat{s}_{\text{mean}})/(s_{\text{var}}/s_{\text{mean}})))^2 = (\log(\hat{s}_{\text{var}}/s_{\text{var}}) - \log(s_{\text{mean}}/\hat{s}_{\text{mean}}))^2$. We chose to use the

Fano factor because it is particularly well-suited for quantifying transcriptional noise in systems where the mean expression level varies across conditions or parameters. Unlike variance, the Fano factor normalizes variability by the mean, making it more robust for comparing noise levels across genes or regulatory regimes with different expression levels. Compared to the coefficient of variation (CV), which normalizes by the square of the mean, the Fano factor tends to be less sensitive to low expression regimes and is commonly used in stochastic gene expression studies, especially when the distribution is skewed or over dispersed (i.e., variance exceeds the mean). This makes it a more appropriate metric in our context, where transcriptional bursting often leads to over dispersed expression distributions. We have added an explanation in the methods 1.3 of revised manuscript to explain this choice.

On page 17, the definition of "sample" is unclear. Does it refer to the number of parameters sets or to the simulated trajectories generated by stochastic simulation algorithms?

Thank reviewers for your valuable feedback. The term "sample" in this context refers to the data points used in the neural network training set. To eliminate any ambiguity, we included a precise mathematical definition of "sample" ($\theta_i, P_{\text{simulation},i}$) in the methods 1.3 section of revised manuscript.

Additionally, it is unclear how the authors determined the number of simulated trajectories per parameter set to ensure training accuracy. Furthermore, it would be relevant to address whether including moments during neural network training is beneficial.

We appreciate the reviewer's insightful questions regarding the simulation and training process. To clarify, for each parameter set, we did not simulate multiple trajectories to obtain the corresponding distribution. Instead, we simulated the system for a sufficiently long

period to ensure that the system reached a steady-state distribution. From this steady-state data, we then used interpolation methods to derive the corresponding distribution for each parameter set.

On the other hand, the moments were calculated theoretically without any approximations, providing higher accuracy. By incorporating the moments into the training process, we can effectively mitigate potential biases arising from insufficient sampling of the simulated data. Moreover, our experiments on the synthetic dataset demonstrate that introducing the moments as a loss function significantly enhances the model's performance on the test set (Figure 2E).

What is the intuition behind the choice of α_{cg} ? On page 18, the rationale for setting the sampling probability to 0.5 is unclear. Could this parameter be inferred rather than being preset?

We thank the reviewer for the insightful comment regarding the choice of α_{cg} . We acknowledge that the typical values of this parameter in related literature often fall within a narrower range (e.g., 0.06–0.32) (Zheng GX et al., 2017; Macosko EZ et al., 2015). However, our decision to set α_{cg} was based on a trade-off between sampling efficiency and computational tractability in our specific application context. While it is indeed possible to infer α_{cg} as a learnable parameter, we opted for a fixed value in this work to reduce model complexity and avoid unidentifiability issues. In addition, we conducted inference under different capture efficiencies (0.5, 0.3, and 0.2), and found that the inferred burst size (BS) and burst frequency (BF) remained strongly correlated across these conditions (Supplementary Figure S12). This indicates that variations in capture efficiency do not significantly impact the outcomes of downstream enrichment analyses. Nevertheless, we agree that adaptively learning α_{cg} could be a promising direction, and we plan to explore this in future work. We have added the explanation in methods 1.4 section.

On page 19, the authors employed gradient descent for parameter inference. However, as this method is sensitive to initial values, it is unclear how the starting points were selected.

We sincerely thank the reviewer for highlighting the sensitivity of gradient-based optimization methods to initial values. To address this concern, we adopted a black-box optimization strategy in the form of the adaptive differential evolution (DE) algorithm (Das S & Suganthan PN, 2010) to derive robust initial parameters for the parameter inference. The adaptive DE algorithm enables global exploration across a broad parameter space, thereby reducing the risk of convergence to suboptimal local minima. This yielded reasonably good initial estimates, which were subsequently refined using gradient-based optimization to identify high-quality solutions characterized by a vanishing gradient norm. This hybrid strategy, which combines global and local search, is widely adopted in optimization literature to alleviate the risk of entrapment in local optima (Ahandani MA et al., 2014). We have clarified this detail in the third result of the revised manuscript.

Furthermore, clarification on how the gradients were computed - whether through finite difference approximation or other methods - would offer additional insight into the robustness and accuracy of their approach.

Thank reviewers for valuable feedback. Regarding the computation of gradients, we use the chain rule in neural networks, and the gradients are computed through backpropagation. Specifically, we rely on automatic differentiation to efficiently calculate the gradients. Unlike finite difference approximation, automatic differentiation directly computes the derivative of the loss function with respect to each parameter, ensuring accurate gradient calculations (Baydin AG et al., 2018). We have clarified this detail in the discussion section of the revised manuscript.

The paper presents several comparisons between continuous and discrete distributions in Figure 2B and Supplementary Figures S4, S6, and S8, described as a "comparison between mRNA distribution and inferred distribution by DeepTX for scRNA-seq data" or a "comparison between SSA results and DeepTX prediction results." This may lead to confusion for the reader, as the paper focuses on transcriptional bursting, a process where we would typically expect the distributions to be discrete. Clarifying this point would help align the figures with the main topic and enhance the reader's understanding.

We sincerely thank the reviewer for this insightful comment. We understand the concern that the distributions shown in Figure 2B and Supplementary Figures S4, S6, and S8 may appear to be continuous, which could be confusing given that transcriptional bursting naturally results in discrete mRNA count distributions.

We have clarified that in all these figures, both the empirical mRNA distributions derived from scRNAseq data and the model-predicted distributions from DeepTX are inherently discrete. To visualize the empirical distributions, we used histograms where the x-axis corresponds to discrete mRNA copy numbers and the y-axis represents the normalized frequency (density). To illustrate the DeepTX-inferred probability mass function, we plotted the predicted probabilities at each integer count as points and connected them with lines for clarity. While the connecting lines give the appearance of continuity, this is a standard graphical convention used to better show trends and model fit in discrete distributions.

We suggest that Figure 3E could present the error as a percentage of the parameter value, as this would provide a more equitable comparison and better illustrate the relative accuracy of the parameter estimation.

Thank reviewers for suggestion regarding Figure 3E. We agree that presenting the error as a percentage of the parameter value would offer a more equitable basis for comparison and better highlight the relative accuracy of our parameter estimation. Accordingly, we have revised Figure 3E to include the relative percentage error for each parameter.

Figure 4A could be improved for better legibility. The contour plots are somewhat confusing, and the light blue points are difficult to distinguish. Additionally, the x-axis label "Untreatment" appears throughout the manuscript-could this term be referring to the control experiment?

Thank reviewers for constructive feedback. We have revised Figure 4A to improve its clarity and legibility. Specifically, we adjusted the display style of the contour plots and enhanced the visibility of the light green points to make them more distinguishable.

Additionally, we recognize the potential confusion caused by the term "Untreatment" and have replaced it with "Control" throughout the revised manuscript to ensure consistency and accuracy in terminology.

Figure 4B was unclear, and further explanation would be helpful for understanding its purpose.

Thank reviewers for feedback. The purpose of Figure 4B is to illustrate the relationship between bursting kinetics and the mean and variance of the model. In the revised manuscript, we will provide a more detailed explanation of how the figure captures these relationships, highlighting the key insights it offers into the underlying dynamics.

Figure 4B illustrates the quantitative relationships among BS, BF, and gene expression noise within the framework of the transcriptional model. In this log-log-log 3D space, the mean expression level is constrained on a blue plane defined by the equation $\log(\text{BS}) + \log(\text{BF}) = \log(\text{Mean})$, highlighting that the product of burst size and burst frequency determines the mean expression level. The orange plane represents a scaling relationship between expression noise and burst kinetics, expressed as $\log(\text{BS}) + \log(\text{BF}) = k \log(\text{Noise})$, where k is a constant indicating how the burst kinetics co-vary with noise. Notably, the trajectory of the green sphere demonstrates that, under a fixed mean expression level (i.e., remaining on the blue plane), an increase in gene expression noise arises primarily from an increase in burst size. We have revised the caption of Figure 4B.

In Figure 4D, two of the GO analysis terms are highlighted in red, but the meaning behind this emphasis is not clear. The same question applies to Figure 5E, where the green dots are missing from the plot.

Clarification on these points would enhance the overall clarity.

We appreciate the reviewer's thoughtful comments. We have added further clarification regarding the enrichment analysis results presented in Figure 4D. Specifically, we highlighted the "cell cycle G2/M phase transition" pathway because a delay in the G2/M phase transition has been shown to increase the probability of cell differentiation, which is a key aspect of our study. In addition, since IdU treatment is known to induce DNA damage, we emphasized the DNA damage-related pathway to support the biological relevance and consistency of our enrichment results. Similarly, in Figure 5E, we highlighted the apoptosis-related pathway. Apoptosis in this context is closely associated with cellular responses to toxic substances and mitochondrial dynamics. The enrichment of pathways related to these processes enables us to hypothesize the underlying mechanisms driving apoptosis in our system. Further, the absence of green dots in Figure 5E was due to an error in the figure caption. We have revised the figure caption accordingly to accurately describe all elements presented in the figure.

Clarify axis labels in figures, particularly the y-axis in Figure 5A and the x-axis in Figure 6G. In the first case, it isn't clear what this "value" represents. In the second case, the x-label is very confusing. As I understand the figure description, in these plots you are always comparing the G0 arrested genes between control and treated cells. But the x-label says "G0 (0 D)", "Cycle (50 D)".

Thank reviewers for pointing out the issues with the axis labels. We have made the necessary revisions to eliminate any confusion. In Figure 5A, the label for the y-axis has been changed from "value" to "log₂ (value)" for clarity. The "value" in y-axis represents the value of statistical measure indicated at top of each panel. In Figure 6G, the x-axis label "Cycle (50 D)" has been updated to "G0 (50 D)" to accurately reflect the comparison between the G0-arrested genes in control and treated cells. We have revised the text of Figure 5A and Figure 6G.

Figure 6 uses a QS metric (quality score), but the definition of this metric is not provided. Including a brief explanation of its meaning would be helpful for clarity.

Thank reviewers for feedback. In this version, we provided explanation of the QS (Quality Score) metric in the supplementary text 3 for better clarity. The QS is calculated based on the difference in z-scores derived from GSVA (Gene set variation analysis) of gene sets upregulated and downregulated during the quiescent phase, and is defined as $QS = z(\text{up genes}) - z(\text{down genes})$, as described in the literature (Wiecek AJ et al., 2023). $z(\text{up genes})$ represents the standardized enrichment score of the gene set upregulated during quiescence in each sample. A higher value indicates that the quiescence-associated upregulated genes are actively expressed, suggesting that the sample is more likely to be in a quiescent (G0) state.

z (down genes) corresponds to the standardized enrichment score of genes downregulated during quiescence. A lower value implies effective suppression of these genes, which is also consistent with quiescence. The difference score QS serves as an integrated indicator of the quiescent state: A higher value reflects simultaneous activation of quiescence-associated upregulated genes and repression of downregulated genes, indicating a gene expression profile that strongly aligns with the G0/quiescent state. A lower or negative value suggests a deviation from the quiescent signature, potentially reflecting a proliferative state or failure to enter quiescence.

In Figure 6G, light grey lines are shown, but their significance is unclear. It would be useful to specify what these lines represent.

Thank reviewers for observation. In Figure 6G, each point represents a single gene, and the light grey lines indicate the trend of changes in the corresponding bursting kinetics values, mean and variance for genes. We have added the explanation in the caption of Figure 6G.

Additionally, the manuscript should include references to the specific pathways used in the GO analysis to provide more context for the reader.

Thank reviewers for the suggestion. We have included references to the specific pathways used in the GO analysis in the revised manuscript to provide additional context for the readers.

In the discussion, sentences like "IdU drug treatment-induced BS enhancement delays the cell mitosis phase transition, impacting cell reprogramming and differentiation" are problematic as they imply causality, which I believe cannot be determined through the present analysis. The strength of the conclusions needs to be better argued (or toned down).

We acknowledge that the original sentence lacked precision and may have conveyed a misleading implication of causality not fully supported by our current analysis. In the discussion section of revised manuscript, we have rephrased the statement to present a more nuanced interpretation: IdU drug treatment-induced BS enhancement of genes may be associated with a delayed transition in the cell mitosis phase, which could potentially influence cell reprogramming and differentiation.

Other (minor) comments:

On pp. 10, "the BS down-regulates differential genes were mainly enriched..." appears to have a grammatical error/typo, "down-regulated"?

We have made correction. We have revised "down-regulates" to "down-regulated" for grammatical consistency.

Equation 2 doesn't match Figure 1A.

We have made correction. The definition of $BF = 1 / (\langle \tau_{off} \rangle + \langle \tau_{on} \rangle)$ in Equation 2 is correct. We have revised the definition of BF in Figure 1A to ensure consistency with Equation 2.

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